

Neural modelling of complex pulse train stimulation to improve cochlear implant coding strategies

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And to my family, for bringing me up, supporting me wherever I was in this huge world, and for their endless love and care

Abstract

Owing to the invention of cochlear implants, ever since 1970-ies it has been possible to restore hearing to patients with severe to profound hearing loss, which is achieved by placing an electrode array in the cochlea and activating the auditory neuron fibers using an electrical stimulus. For a given sound reaching the implant (its external parts), the appropriate temporal pattern of electrical stimulus for each electrode is calculated using dedicated coding strategies. Due to the spread of the electrical stimulus across electrodes and nonlinear temporal response of neurons to stimulus, both neglected by typical coding strategies, the resulting neural excitation only roughly represent the applied stimulus, which reduced the quality of hearing for patients, especially regarding to temporal fine structure (i.e. music) or sound in noise scenarios.

Novel coding strategies which aim to alleviate these shortcomings by taking considering the neural dynamics (e.g. Refractory State Coding [Babacan 2010]) are being developed, but testing these coding strategies clinically is time consuming for both researchers and patients. A convenient alternative would be to develop a neuronal model capable of predicting the response of auditory neuron fibers to the applied stimulus, thus enabling researchers to quickly test if newly developed strategies could bring improvement in patient's hearing.

To this end, in this study a single neuron phenomenological neural model by [Boulet et al 2018] has been extended to a population of neurons, and the effects of spread of excitation, variability in refractory time scales have been introduced. To enable easier testing of model predictions for potential users, a graphical user interface has been developed which allows arbitrary combinations of the considered neural phenomena and modifications of all relevant parameters. Additionally, to enable comparison of model predictions to the data obtained using neural response telemetry (NRT), a model of the evoked compound action potential (ECAP) was implemented and used to predict the ECAP responses based on model-obtained spike histogram. From the computational side, implementation of this code shows a reduced computational load compared to a similar model by [Tabibi et al 2018], and further modifications which can lead to further acceleration of code runtime have been.

Since the previous models were able to predict responses to constant amplitude pulse trains, focus of this study has been put on the more realistic scenario of amplitude modulated pulse trains, for which previous models have shown limited predictive power [Hess 2017]. Our model has been able to improve on the previous studies, matching the general behavior of ECAP responses to various modulated pulse trains (e.g. phase shift and reduction in width for sinusoidal modulation), and the spike rate adaptation has been identified as a dominant effect shaping the ECAP response in such cases.

A key deficiency has been found to be the simplistic ECAP model used in our model, which due to assumption of no latency predicts smaller modulation depths in ECAP responses. Here this shortcoming has been performed in an „ad hoc“ manner, where stimulus modulation depth in the model was artificially increased when fitting the data. This approach has produced satisfactory results, but in the future work the ECAP model needs to be improved to enabled proper comparison of predicted and measured ECAP responses, and thus proper validation and usage of our model.

Measurements needed to extract the neural parameters needed for our model have been suggested, and due to a limited set of data used in this study it is expected that such measurements will provide a more suitable test of our model. Together with other improvements suggested in this study, this will hopefully help reach a verdict on the usability of our model for predicting responses of neural populations to stimuli generated by novel coding strategies aimed at improving hearing for cochlear implant patients.

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1. Introduction

The building block of any society is the ability to communicate, to share and discuss one's ideas, opinions and emotions to fellow beings, and thus be included in that society. Two major communications modalities humans use are the visual and auditory modalities, where the auditory modality (mostly speech) is typically synonymous with direct and face-to-face communication with another person. Losing any of these modalities will create huge obstacles in taking part in the society – and in the case of the hearing loss, this means not being able to hear your alarm clock in the morning, your family calling you to dinner, enjoying a conversation in a busy restaurant, or just enjoying the subtle sounds of nature. Luckily, with the invention of cochlear implants, ever since 1970-ies it has been possible to restore hearing even to persons with severe to profound hearing loss ("deaf" persons), and today there are more than 400.000 patients who are part of the "hearing world" thanks to cochlear implants.

1.1. Acoustic hearing

To understand why cochlear implants function in the manner they do, one should first understand the basics of natural (acoustic) hearing, i.e. the path from acoustic stimulus to sound perception in the brain. First, the acoustic stimulus is gathered by the outer ear and then focused down the ear channel onto the eardrum, thus providing first step of amplification. Then, vibrations of the eardrum cause the motion of tiny bones in the middle ear (hammer, anvil and stirrups), which act like a system of levers amplifying and transferring the incoming stimulus to the oval window, a flexible membrane on the inner ear.

Movement of the oval window then creates an acoustic wave which propagates in the cochlea, a spiral hollow chamber in the inner ear separated by a (basilar) membrane into two sections. This acoustic wave first propagates up along one of the scales (*scala vestibuli*), reaches the junction of the two scales (*helicotrema*), and then propagates down the other scale (*scala tympani*) and reaches the round window, where reflection of the incoming wave is avoided since the round window is movable.

Since a propagating wave will cause a difference in pressures on two sides of the basilar membrane, this causes movement of the membrane at a frequency matching the wave's frequency. However, since the basilar membrane is stiffer at the base (closer to oval window) than at the apex (near *helicotrema*), deflection of the membrane will be strongest at the location where the membrane eigenfrequency matches the wave frequency. Hence, incoming sounds of different frequencies will "excite" different parts of the membrane, thus providing a 1:1 mapping from sound frequency to stimulus location. This mapping is commonly known as the tonotopic principle and is crucial to functioning of the cochlear implant.

From here on, the mechanical movement of the basilar membrane is transduced into an electrochemical signal. Such a crucial feat is performed by the inner hair cells (IHC) which become deflected during membrane movement, which in turn causes potassium ion channels located on the inner hair cells to open, leading to depolarization and through synaptic connection to excitation of auditory nerve fibers. This marks the transition of the stimulus to the peripheral neuronal system, and from here on the neuronal signals are further processed in the higher levels of auditory system, until they reach the brain where the sound perception finally forms.

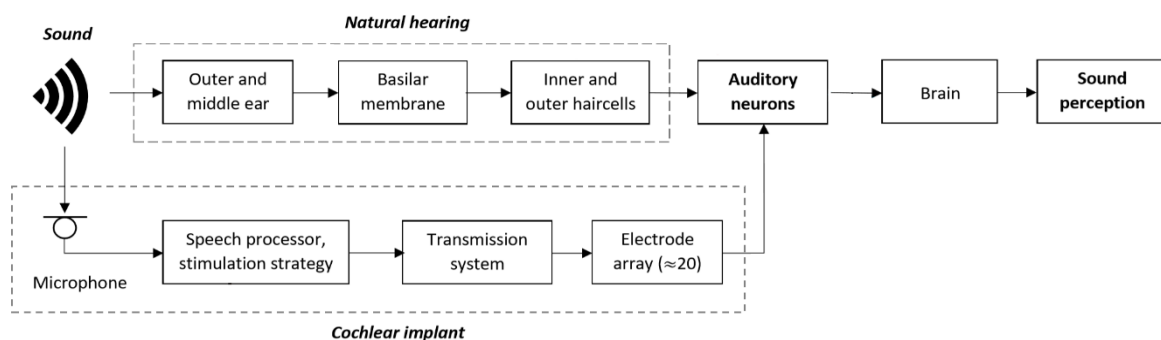
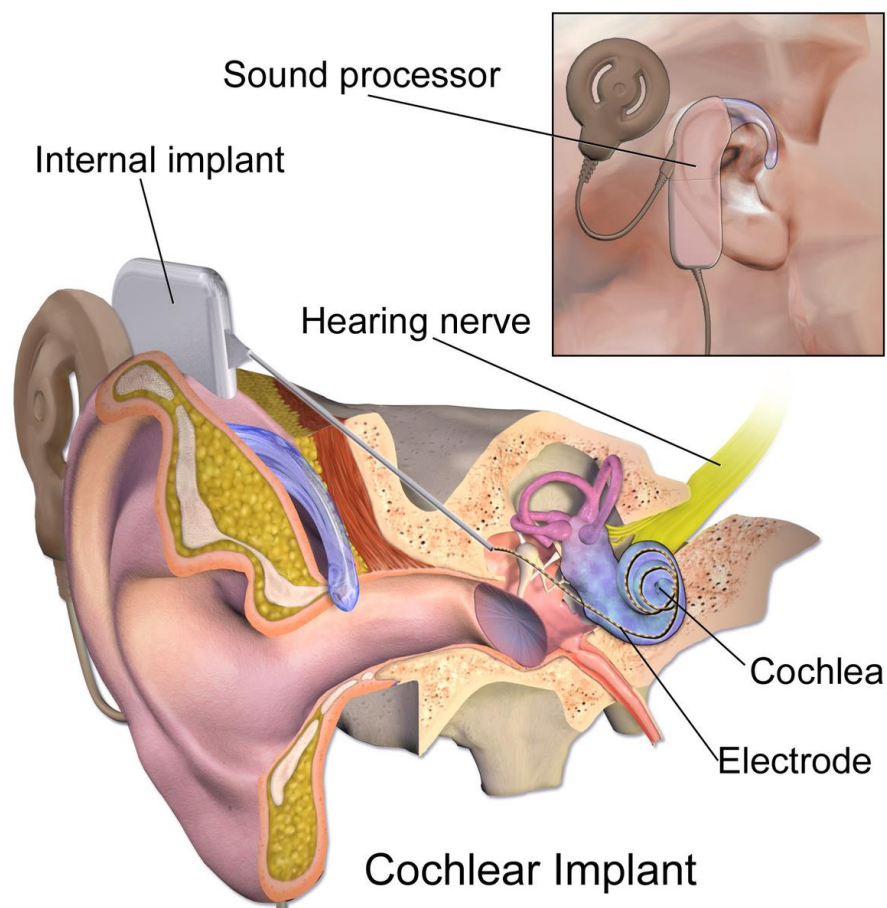


Figure 1.1.: Overview of natural hearing and cochlear implant hearing. Adapted from [Hamacher 2004].

1.2. Electrical hearing

In the majority of cases, the underlying cause of hearing loss is the damage or loss of (inner) hair cells [Loizou 1999], which is in turn caused by exposure to loud sounds, aging, ototoxic chemicals *etc.* The reason why the loss of inner hair cells causes hearing loss should be clear from the previous section: inner hair cells perform the crucial role of transducing the mechanical stimulus to an electrochemical signal, and without them the auditory nerve fibers cannot be excited, and thus no sound can be heard. A method to restore hearing in such case would be to directly stimulate the auditory neurons and hopefully create a meaningful sound perception – which is exactly what cochlear implants accomplish.

Cochlear implants (CI) have existed for a relatively long period, but the basic parts (shown in Fig. 1.1.) have not changed: a cochlear implant consists of an electrode array, which is placed into the cochlea, internal and external coils which are used to transmit power and stimulus to the electrode array, and a sound processor which records the incoming sound and determines the appropriate electrical stimulus to be send to the electrodes. Likewise, the basic principles behind functioning of cochlear implants have remained the same: since the ionic solutions in cochlea are conductive, the applied stimulus will spread from the electrodes to the auditory neuron fibers, and due to the tonotopic organization of the cochlea, stimulus at each electrode will be perceived as a sound of a different pitch. Together with the inherent plasticity of the human brain, this enables a cochlear implant patient to, given enough time and practice, adjust to this new way of hearing.



*Figure 1.2.: Positioning and components of a cochlear implant.
Source: Blausen.com staff, Medical gallery of Blausen Medical, 2014, WikiJournal of Medicine 1 (2), Creative Commons license*

At this point, one might put forward two important questions about how cochlear implants work: how does one determine an appropriate electrical stimulus based on the recorded sound, and how do the auditory neuronal fibers respond to the applied electrical stimulus? The first question refers to what is called the cochlear implant coding strategy, and while many coding strategies have been developed, one of the typical answers is to:

1. Separate the incoming sound into n frequency bands
2. Extract the sound envelope for each frequency band
3. Generate electrical stimulation proportional to the envelope amplitude, either for each frequency band, or m out of n bands with the highest envelope amplitudes
4. Send the signal to the corresponding electrodes (i.e. positions along the basilar membrane)

The main reason that only the sound envelope is considered, is that this is sufficient to restore speech intelligibility (i.e. communication), which is the primary goal of any cochlear implants. However, this also means that the fine temporal structure of sound necessary to appreciate music or understand tonal languages (e.g. Mandarin) is degraded, and similarly, pitch perception and intelligibility of speech in noise („cocktail party“ scenario) are limited [Schnupp *et al* 2012].

Similarly, yet another issue with current coding strategies is that they typically neglect the second question from above, i.e. they neglect the highly nonlinear and stimulus-dependent response of neurons to the applied electrical stimulus, which produces a response which can greatly vary from the desired response. For example, neurons show a characteristic behavior of refractoriness, a reduction in excitability after spiking, which means that attempting to stimulate neurons shortly after they have spiked is an unnecessary waste of energy. In order to alleviate these shortcomings new coding strategies are being developed which do take into account the neural dynamics (e.g. the Refractory State Coding by [Babacan 2010]), but testing these coding strategies clinically is time consuming for both researchers and patients. A convenient alternative would be to develop a neuronal model capable of predicting the response of auditory neuron fibers to the applied stimulus, thus enabling researchers to quickly test if newly developed strategies could bring improvement in patient's hearing.

Modelling the behavior of neurons has been a long lasting interest of scientists, and in general the approaches taken so far can be classified into two categories. Firstly, one can adopt a bottom-top approach where neural behavior is modelled at the microscopic level, and then use this knowledge as a building block for deriving a whole-neuron model. Such biophysical models offer great precision and understanding of neural behavior, with the most famous example being the Hodgkin-Huxley model (describing a single node of a neuron), which further extended with a cable model can model a non-myelinated neuronal fiber. However, such models are computationally intensive, and from an experimentalist or clinical point of view, changing parameters in such models does not lead to easily predictable changes in the neural response, and they are hence hard to relate to the data.

As an alternative, a second approach is to model neuronal behavior from a top-bottom perspective, aiming only to replicate the general phenomenology of their behavior. A well-known example is the integrate and fire model where the whole neuron is modelled as a leaky capacitor, and when the neuron is charged above a certain voltage threshold, a spike is produced. With the addition of stochasticity and refractory behavior such models have become much better at predicting neural response to electrical stimulus [Seeber & Bruce 2016], and compared to the biophysical models they have the advantage of lower computational complexity and more easily recognizable relation of model parameters to the data. Predicting responses to high-rate stimuli or pulse trains has though remained a challenge, but a recently developed single-neuron model [Boulet 2018], which for the first time included the additional neural behaviors of spike-rate adaptation, facilitation and accommodation, has enabled better prediction of responses in such cases and for unmodulated pulse trains. However, this model considered only a single neuronal node, and hence a natural extension would be to consider a population of neurons, and quite importantly, responses to speech-like modulated pulse trains.

Development of such an extended model is the main goal of this project, and the following sections of this thesis will introduce the basic phenomena of neural behavior, their implementation in a computational model for a population of neurons, experimental methods needed to collect the necessary data for parameter fitting and model validation, and finally the obtained results and their discussion.

2. Neural phenomena

To understand how a group of neurons respond to an electrical stimulus, a logical first step is to understand how a single neuron responds to electrical stimulus. Even though it was known for a long time that living beings (tissues) respond to electrical stimuli – with the most famous example being the Galvani experiment with frog legs – it was only in 1952 that [Hodgkin and Huxley 1952] had the tools and knowledge to successfully explain the many peculiarities of single neuron behavior. These include for example: sharp cutoff between neuron spiking or not spiking, reduced excitability following a spike, increased and then decreased excitability following a non-spiking response, the time scales of these phenomena, and many more. In the following sections we shall introduce each of these characteristic neuronal behaviors, with the addition of phenomena not captured in the Hodgkin-Huxley model, and phenomena important for extracellular stimulation of a neuronal population.

2.1. Neuronal membrane

Early in the 20th century, [Lapicque 1907] had conducted a series of measurements where strength and duration of electrical stimulus were varied and it was observed if neurons would produce a spike. As it was suggested by this data, a neuronal membrane can be modelled as a leaky capacitor (Fig. 2.1.), for which we can write:

$$I_{stim}(t) = I_c(t) + I_R(t) = C \frac{dv}{dt} + \frac{v(t) - v_{rest}}{R} \quad (1)$$

If we now consider only the change in the membrane voltage relative to the rest voltage, we obtain the following ordinary differential equation:

$$\frac{dv(t)}{dt} = -\frac{v(t)}{RC} + \frac{I_{stim}(t)}{C} \quad (2)$$

which for simple types of stimulus has a familiar exponentially rising and falling solution with a time scale of $\tau = RC$, typically on the order of 0.1 ms.

The interpretation of this equation is the following: during stimulus, the neuronal membrane will charge up and reach a saturation point if the stimulus is longer than the membrane time scale, and when the stimulus is not applied, the membrane will discharge with the membrane timescale (Fig. 2.2.). In addition, if a weak stimulus which does not cause a spiking response is followed by another weak stimulus, but with a delay smaller than the membrane timescale, membrane responses to these stimuli will sum up, and can together cause a spiking response. This summation or integration of stimuli is known as passive facilitation, and it has an important role in increasing the excitability of neurons and enabling even the (quick) sub-threshold pulses to cause a spiking response.

If we now take that the membrane voltage is updated at each timestep according to equation 1 and that the neuron spikes (i.e. spike timing is saved and voltage reset) whenever the voltage reaches a certain threshold, we arrive to a well-known integrate and fire model. While being the simplest neuronal model in existence, for low frequencies it can predict spike timings relatively well (which is the most important information that can be obtained), and more importantly, it is our first building block towards a better model.

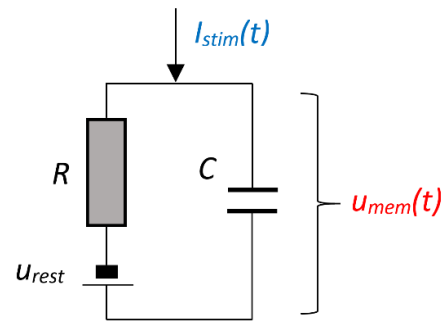


Figure 2.1.: Electrical circuit equivalent of a neural membrane.

Adapted from Gerstner W. Et al, Neuronal Dynamics, Cambridge University Press

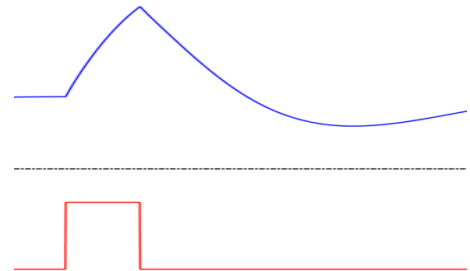


Figure 2.2.: Membrane voltage response to a subthreshold stimulus, obtained from a simulation of a canonical HH model, showing the charging and discharging behavior of the neural membrane. Blue: neural membrane voltage, Red: stimulus pulse, Dashed black: threshold stimulus for causing a spiking response

2.2. Neural membrane noise

Since the model described above is deterministic, producing identically ordered spike times for identical stimuli, which is contrary to the observed stochastic firing behavior of neurons, the next simple modification would be to consider the influence of noise.

As one could expect, neurons as a biological system are prone to noise generated by various sources: random fluctuations in ion concentrations on either side of the membrane, changes in conformation of ion channels (and thus their conductivity), shot (“granular”) noise due to discrete nature of ion charge (leading to discrete changes in membrane voltage), *etc.* Together, these noise sources produce a noise of $1/f$ spectrum and a Gaussian amplitude distribution, as it was first reported by [Verveen and Derkens 1968].

In order to describe the effect of noise mathematically, one can observe the firing probability of a neuron versus the stimulus level. Namely, in the absence of noise this dependence can be described with a step function, but with the addition of noise there is a probability of neuron firing even when below resting potential, and likewise a probability a neuron will not fire even when above the resting potential [fig 2.3.]. Mathematically this corresponds to a cumulative (integrated) Gaussian function of width proportional to the membrane noise, which therefore enables one to calculate the noise level. Typically, the noise level is reported as a relative spread (RS), i.e. as a quantity scaled relative to the (mean) current threshold:

$$RS = \frac{\sigma}{\theta} \quad (3)$$

where θ is the (mean) stimulus current threshold and σ is the (calculated) threshold spread. From here on, the equivalent noise level of the membrane voltage can be calculated and inserted into the integrate and fire model, thus introducing stochastic behavior to the model as desired.

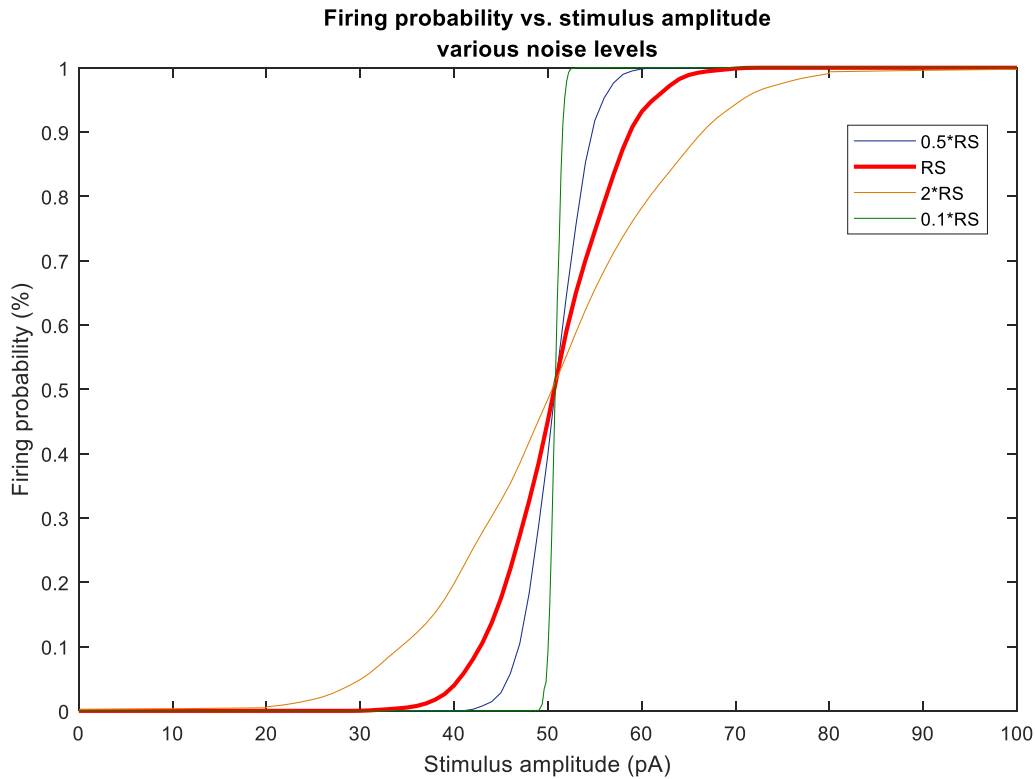


Figure 2.3. Single neuron firing probability versus single pulse stimulus amplitude for various noise relative spreads, obtained with our computational model. Without any noise the threshold stimulus current is 50 pA, but with presence of noise the transition from non-firing to firing behavior becomes less sharp, with width proportional to the relative spread, and the zero-noise threshold now corresponds to 50% firing probability level. All relative spreads above are reported relative to the value of relative spread used in our model (Table 1).

2.3. Refractoriness

The most well-known nonlinear behavior characteristic to neurons is refractoriness. From a phenomenological point of view, refractoriness is manifested as reduction of neuron's excitability after a spiking response: for a short time period after the spike, called absolute refractory period (ARP), the neuron will not be excitable regardless of the stimulus strength. Later on, during the relative refractory period (RRP), the neuron will slowly return to its rest excitability and will be able to produce a spike when given a stronger stimulus than required in the rest state (Fig. 2.4.). Microscopically, the cause of refractory behavior is the inactivation of Na^+ ion channels: these channels are responsible for producing a spiking response, and following a spike they return to their "rest" state (de-inactivate) only after a period lasting on the order of 1 ms or lower.

Such behavior is predicted by the canonical Hodgkin-Huxley model, and using a stimulation with two pulses (masker-pulse) with the first pulse causing a spiking response, one can obtain the relation of the interval between two pulses (masker-pulse interval, MPI) and the pulse strength necessary to cause a spike during (relative) refractory period.

Specifically for the case of auditory neuron fibers, additional ion channels not considered in the Hodgkin-Huxley model play an important role. As was shown by [Negm and Bruce 2011], both ARP and RRP durations are prolonged by the presence of KLT channels (potassium low threshold) and HCN channels (heterocyclic cation nucleotide). Moreover, since the number of such channels varies from one Ranvier's node to the other, ARP and RRP duration are variable across different neurons (and hence between patients and electrodes), as it was shown in measurements performed on feline auditory neuron fibers by [Miller et al 1999], and such variation can play an important role in shaping the neuronal population response.

For example, for a single neuron the refractory behavior limits the maximum possible firing frequency (theoretically to the inverse of ARP duration), but given a population of neurons and variations in ARP/RRP duration, it is possible for neurons to take turns in firing when stimulated by a high-frequency stimulus (volley principle), albeit with only a part of the neuronal population firing during each pulse [Schnupp et al 2012]. Similarly, depending on the level of ARP/RRP variability between neurons, a majority of neurons could fire in response to the first stimulus pulse, leaving only a handful of neurons ready to fire at the second stimulus pulse. Depending on the value of (mean) RRP and the stimulus frequency, this cycle will repeat itself and thus produce an oscillatory response to a constant pulse train stimulus, with the amplitude of oscillations being lower for larger ARP/RRP variations. Along the same line of reasoning, given that the initial pulse of a high-frequency pulse train is strong enough to cause spiking in the majority of neurons, variations in ARP/RRP will make only a part of neuronal population excitable at each pulse, hence making the long-term response noticeably lower as compared to the initial response [Rubinstein et al 1999].

Due to its strong effect on the neural response and for being a well-known and characterized neural behavior, refractoriness has been included in many (phenomenological) neural models to date [Seeber and Bruce 2016], including our model. Even though such models are much improved as compared to the "bare" integrate and fire model described earlier, they still do not capture the full spread of neural behaviors, and hence additional neural phenomena need to be considered.

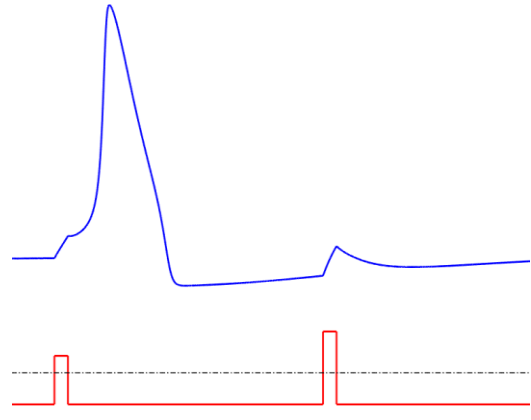


Figure 2.4.: Membrane voltage response to two closely timed suprathreshold stimuli, obtained from a simulation of a canonical HH model, showing the effect of refractoriness. Blue: neural membrane voltage, Red: stimulus pulse, Dashed black: threshold stimulus for causing a spiking response

2.4. Spike rate adaptation

Besides refractoriness, another spike-dependent behavior to be included in our model is the spike rate adaptation. As was noted in [Boulet et al 2018], “spiking neurons pay a substantial metabolic cost to encode information about the environment”, and hence, from an evolutionary point of view one might expect that neurons will tend to reduce their firing rate over time, i.e. to show adaptation effects, as is regularly observed in neural responses [Woo et al 2009].

Microscopically, such an adaptation can happen due to increase in extracellular concentration of K^+ ions (due to outflux of K^+ ions during each spike), as was shown in a model by [Woo et al 2009], and the general effect of spike rate adaptation is to increase the firing threshold by a certain value whenever a neuron spikes, with this increase exponentially diminishing with time. Together with the effects to be mentioned next, spike rate adaptation (SRA) plays an important role in decreasing the neurons excitability during its ongoing stimulation such as (un)modulated pulse trains, and hence will be included in our model.

2.5. Active facilitation and accommodation

Until now, we have put our attention to the influence of spike-dependent behaviors on the firing threshold. However, subthreshold stimuli can also influence the firing threshold, either increasing or decreasing it, and here we shall mention two effects which are stimulus-dependent phenomena.

Already when we discussed the response of a neural membrane to an electrical stimulus, we have noted the effect of passive facilitation, i.e. the summation of membrane responses to two subthreshold stimuli, which for a small enough interval between pulses cause a spiking response. In addition to this passive effect, the Hodgkin-Huxley model also predicts an active type of facilitation, where due to the activation of Na^+ channels during the first pulse (hence increased membrane conductivity to Na^+ ions) the membrane response to the second pulse is slightly stronger than to the first pulse, thus further increasing the summed neural membrane response (Fig. 2.5.)

This effect however has a limited duration due to inactivation of Na^+ channels (on the time scale of ca. 0.5 ms), and is followed by an opposite effect called accommodation. Namely, following a subthreshold stimulus, Na^+ ion channels will become activated (facilitation), but at time scales larger than 0.5 ms Na^+ ion channel de-activation and inactivation will cause an “overshoot” and thus a reduced excitability compared to the neuron’s rest state (Fig. 2.6.). Additionally, for auditory neuron fibers, presence of HCN channels which activate during a stimulus [Boulet 2016] will further reduce the neuron’s excitability.

Effects of facilitation and accommodation can be characterized in a similar fashion as refractoriness - by applying two pulses (masker and probe) while varying the masker-probe intervals (IPI) and observing if a spike is produced - with the important distinction that the masker pulse is below threshold i.e. masker does not cause a spiking response. Using such an approach applied to the canonical Hodgkin-Huxley model, a curve such as shown in (Fig. 2.7.) can be obtained, where two distinct regions where either facilitation (both passive and active) or accommodation dominate in influencing the neuron’s excitability.

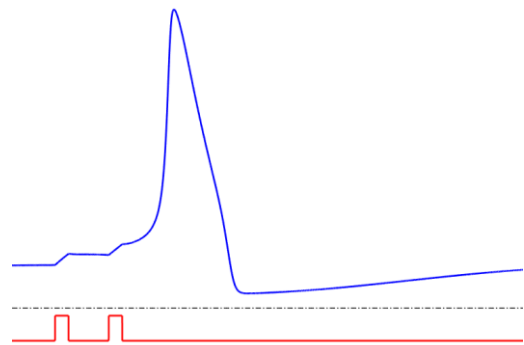


Figure 2.5.: Membrane voltage response to two closely timed subthreshold stimuli, obtained from a simulation of a canonical HH model, showing the effect of facilitation. Blue: neural membrane voltage, Red: stimulus pulse, Dashed black: threshold stimulus for causing a spiking response

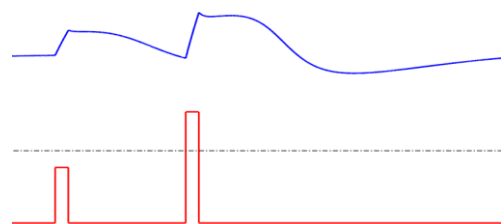


Figure 2.6.: Membrane voltage response to two closely timed subthreshold stimuli, obtained from a simulation of a canonical HH model, showing the effect of accommodation. Blue: neural membrane voltage, Red: stimulus pulse, Dashed black: threshold stimulus for causing a spiking response

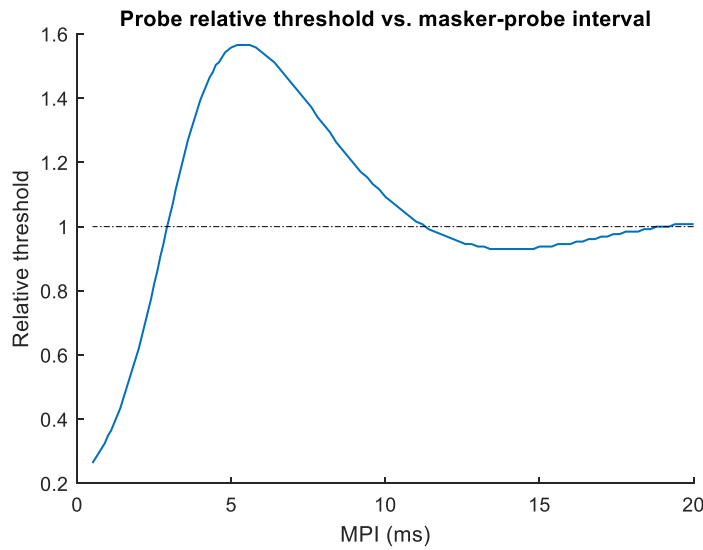


Figure 2.7.: Probe threshold plotted for various masker-probe intervals, simulated using a canonical HH model. Masker stimulus level was set to 80% of single pulse threshold, thus producing no spike, and for each MPI, probe level was varied until a spike was detected. The effects of passive and active facilitation lower the probe threshold during small MPI's, however for larger MPI's it is markedly increased. The following decrease in probe threshold for large MPI's is not observed in auditory neuron fibers and is thus not included in our model [Dynes 1996].

2.6. Spread of excitation

Stimulating auditory neuron fibers using electrical stimuli has one major, but unavoidable drawback: the ionic solution in which electrodes are placed is conductive, and hence the electrical stimulus spreads not only to the auditory neural fibers closest to the electrodes, but also to the nearby fibers. This causes a considerable amount of crosstalk between electrodes, which due to the tonotopic organization of the cochlea implies a reduced frequency resolution for CI patients and thus a reduced quality of sound. Currently this issue can be partly reduced by using a bipolar stimulation (where one electrode acts as a current source and the other as a sink, thus limiting the spread) as opposed to the monopolar stimulation, but in general the spread of excitation cannot be completely avoided and thus needs to be included in any population model.

When aiming to develop a high precision model of spread of excitation, one could make use of the finite elements method (FEM) to calculate the values of electric potential across the cochlea and auditory neuron fibers [Cohen *et al* 2003b], since FEM simulations allow considering complex geometry and heterogeneous media. However, given that our model aims for phenomenological description of all phenomena and low computational loads, a simple model shall be employed, where currents delivered to each fiber are scaled relative to the “center” fiber (the one closest to the stimulating electrode), with the scaling given by an exponentially decaying function [Hamacher 2004]. Given the findings in previous studies [Cohen 2009b], the physical width of the spread is taken to increase with the stimulus level by keeping the mathematical width (i.e. the exponential falloff factor) constant relative to the stimulus level.

CHECK

3. Methods

3.1. Dynamic Threshold potential model

In this section the basics of our computational model will be presented, which represents an extended version of the single fiber computational model by [Boulet et al 2018]. In general, the model is an advanced version of the integrate and fire model, where the threshold voltage and neural membrane noise are modified separately by each of the previously mentioned neuronal phenomena (described as multiplicative factors changing in time) - hence the name “dynamic threshold potential model”. The distinct advantage of such model is that influences of various neural phenomena on the (compound) neural response can be considered individually or in any combination one wants to consider, hence providing a useful tool for researchers.

3.1.1. Model basics: stochastic integrate and fire model

As mentioned previously, this model builds upon the integrate and fire model, hence the membrane voltage is updated at each timestep by solving equation (2):

$$\frac{dv(t)}{dt} = -\frac{v(t)}{RC} + \frac{I_{stim}(t)}{C} \quad (2)$$

and a spike is produced whenever the membrane voltage reaches the (noisy) threshold voltage. Here a spike is “encoded” by resetting the membrane voltage to a predefined reset value and by saving the spike timing (for each spiking neuron), where spike timing is later used both in influencing other neural behaviors (e.g. refractoriness) and for comparing model predictions to the data.

Noisiness in the threshold voltage could be included by adding random noise to the stimulus current, as done in previous studies [ref **CHECK**], but this would be ill-advised in our case since the noise across all neurons would become correlated. Hence, noise is to be added either to the individual neuron’s membrane potential or to its threshold potential, and since (in the first approximation) both choices are equivalent, following the model by [Boulet et al 2018] we shall add the noise to the threshold voltage.

Since all of the neural phenomena we described will modify both the threshold and noise voltage as multiplicative factors, we thus have:

$$v_{\theta}(t) = v_{\theta,mod}(t) + v_{\sigma,mod}(t) \quad (4)$$

where:

$$v_{\theta,mod}(t) = x_{\theta}(t) v_{\theta,0} \quad (5)$$

$$v_{\sigma,mod}(t) = x_{\sigma}(t) v_{\sigma,0}(t) \quad (6)$$

and where:

$$x_{\theta}(t) = x_{\theta,ref}(t) x_{\theta,SRA}(t) x_{\theta,fac}(t) x_{\theta,acc}(t) \quad (7)$$

Since the effect of each neural behavior on threshold noise will be defined through their effect on relative spread, which is normalized with respect to the dynamic threshold potential, we further have:

$$x_{\sigma}(t) = x_{RS}(t) x_{\theta}(t) \quad (8)$$

where:

$$x_{RS}(t) = x_{RS,ref}(t) x_{RS,SRA}(t) x_{RS,fac}(t) x_{RS,acc}(t) \quad (9)$$

Hence, each neural phenomenon is described by its specific threshold x_{θ} and relative spread x_{RS} factor.

In our model implantation, the un-modified threshold voltage noise $v_{\sigma,0}(t)$ is sampled from a Gaussian distribution with a relative spread of $RS = 0.0774$ [Fredelake and Hohmann 2012], i.e. with a standard deviation of:

$$v_{\sigma,0} = RS \ v_{\theta,0} \quad (10)$$

Additionally, thus obtained noise is low-pass filtered with a cutoff frequency 200 Hz [Fredelake and Hohmann 2012], and then rescaled since filtering reduces the noise standard deviation. This step is crucial since sampling the noise at each iteration will produce noise of a frequency range proportional to the inverse of the timestep resolution (i.e. of ever higher frequency with increasing resolution), and hence during each stimulus pulse there is a high probability that at least one of many sampled noise values will be low, thus increasing the neuron's firing probability and overall firing rates. Alternatively, one can also use noise of a fixed spectrum, such as $1/f$ noise with Gaussian amplitude distribution [Verveen and Derkens 1968], or any other noise as long as it has a suitable frequency spectrum and standard deviation.

3.1.2. Refractoriness

Based on the observed refractory/recovery functions [Miller et al 2001], denoting the time since the last spike with Δt_{spike} , we shall model the effect of refractoriness on the threshold voltage as:

$$x_{\theta,ref}(t) = \begin{cases} \infty & , \quad \Delta t_{spike} < T_{ARP} \\ \frac{1}{1 - \exp\left(-\frac{\Delta t_{spike} - T_{ARP}}{\tau_{RRP}}\right)} & , \quad \text{else} \end{cases} \quad (11)$$

i.e. the neuron will be unable to fire during the absolute refractory period T_{ARP} and will return to rest excitability with a time constant τ_{RRP} during relative refractory period, as demonstrated in (Fig. 3.1.) below.

The effect of refractoriness on the noise relative spread during relative refractory period is modelled as an increase in relative spread which exponentially decreases towards its resting value [Miller et al 2001]:

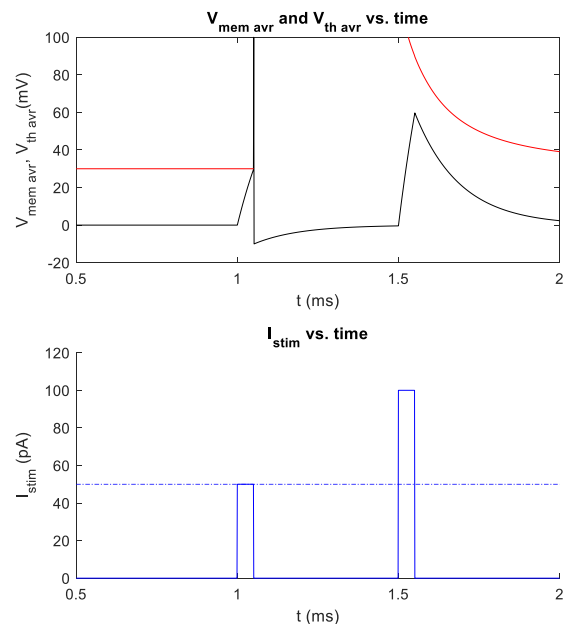
$$x_{RS,ref}(t) = 1 + a_{RS,ref} \exp\left(-\frac{\Delta t_{lastspike} - T_{ARP}}{\tau_{RRP}}\right) \quad (12)$$

where the value of relative spread strength factor $a_{RS,ref}$ and all other parameters are listed in (Table 1)].

Due to the important role variations of ARP/RRP temporal parameters play in shaping the neural population response, these parameters are sampled from a normal distribution of predefined average and standard distribution, and since ion channels responsible for variations in ARP/RRP parameters are co-located [Negm and Bruce 2011], the sampled values are correlated [Hess 2017].

Figure 3.1. Single neuron membrane voltage response and firing threshold versus time for two closely spaced stimuli obtained using our model, showing the modeled effect of refractoriness. All other neural phenomena, including noisiness, were disabled when producing this plot.

Upper plot: black: neural membrane voltage, red: threshold voltage; Lower plot: blue: applied stimulus, dashed black: threshold stimulus for causing a spiking response



3.1.3. Spike rate adaptation

As in the model by [Boulet et al 2018], spike rate adaptation is modelled by increasing the threshold voltage (i.e. the respective multiplicative factor) by a certain value whenever a spiking response occurs:

$$x_{\theta,SRA}(t) = x_{\theta,SRA}(t) + p_{\theta,SRA} \quad (13)$$

where the resting value of the multiplicative factor is unity. Otherwise, the effect of spike rate adaptation undergoes recovery towards its resting value according to the following differential equation:

$$\frac{d}{dt}(x_{\theta,SRA}(t)) = \frac{1}{\tau_{\theta,SRA}}(1 - x_{\theta,SRA}(t)) \quad (14)$$

i.e. it experiences an exponential decrease with a time constant of $\tau_{\theta,SRA}$, as shown in (Fig. 3.2.) below. Due to the same reasons as in [Boulet et al2018], the effect of SRA on relative spread being unknown, it is assumed that SRA affects the relative spread and the threshold in the identical manner.

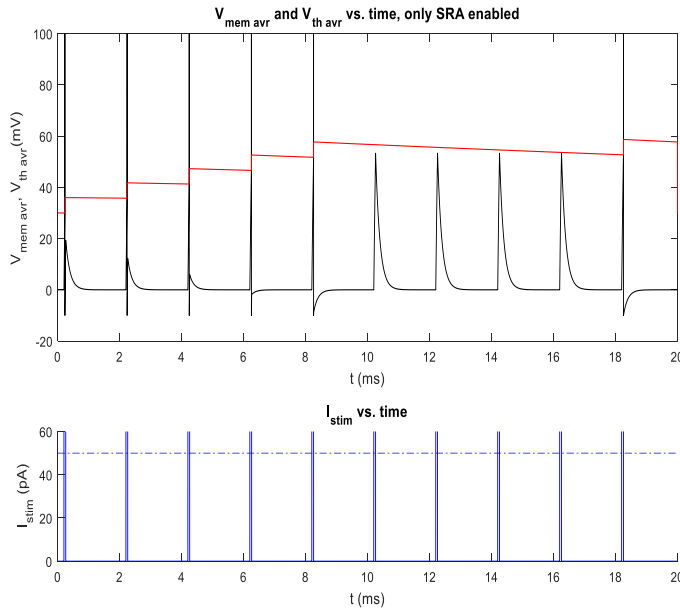


Figure 3.2. Single neuron membrane voltage response and firing threshold versus time for a pulse train obtained using our model, showing the modeled effect of spike rate adaptation. All other neural phenomena, including noisiness and refractoriness, were disabled when producing this plot, hence there is no increase in threshold due to refractoriness following a spike.

Upper plot: black: neural membrane voltage, red: threshold voltage; Lower plot: blue: applied stimulus, dashed black: threshold stimulus for causing a spiking response

3.1.4. Active facilitation and accommodation

Given the dependence of probe relative threshold versus the masker-probe interval (as roughly shown in Fig. 2.6.) and results of previous studies [Dynes et al 1996], the effects of facilitation and accommodation on threshold and relative spread are modelled as in [Boulet et al2018] with a combination of two exponential functions with differing strengths and time constants, where the accommodation time constant is longer than the facilitation time constant. Since facilitation and accommodation both depend on the stimulus level, they are mathematically described by their impulse response functions:

$$h_{\theta/RS,fac/acc}(t) = a_{\theta/RS,fac/acc} \exp\left(-\frac{t}{\tau_{\theta/RS,fac/acc}}\right) \quad (15)$$

and for some normalized membrane voltage $z(t)$ we have:

$$y_{\theta/RS,fac/acc}(t) = h_{\theta/RS,fac/acc}(t) * z(t) \quad (16)$$

where $y_{\theta/RS,fac/acc}(t = 0) = 0$, and where:

$$x_{\theta/RS,fac/acc}(t) = 1 + y_{\theta/RS,fac/acc}(t) \quad (17)$$

Since convolution is numerically costly, taking a derivate of the above expression we arrive at an equivalent but more efficient description using a differential equation:

$$\frac{d}{dt} y_{\theta/RS, fac/acc}(t) = -\frac{1}{\tau_{\theta/RS, fac/acc}} y_{\theta/RS, fac/acc}(t) + a_{\theta/RS, fac/acc} z(t) \quad (18)$$

To avoid „activation“ of facillitation and accommodation while the stimulus is applied, normalized membrane voltage is delayed in time by the pulse width [Boulet et al 2018]., i.e.:

$$z(t) = \frac{v_{mem}(t - t_w)}{v_{\theta,0}} \quad (19)$$

Similarly, since the Na^+ ion channels causing the active facilitation effect are reset after a spiking response and to prevent accumulation of facilitation over two pulses, values of factors $y_{\theta/RS, fac}$ are reset at each spike and at the end of each stimulus pulse, respectively [Boulet et al 2018].

In addition, to enable fitting the effect of weak long term accommodation, as in the previously developed model a “slow” accommodation is added to the model, with identical mathematical behavior but differing parameters.

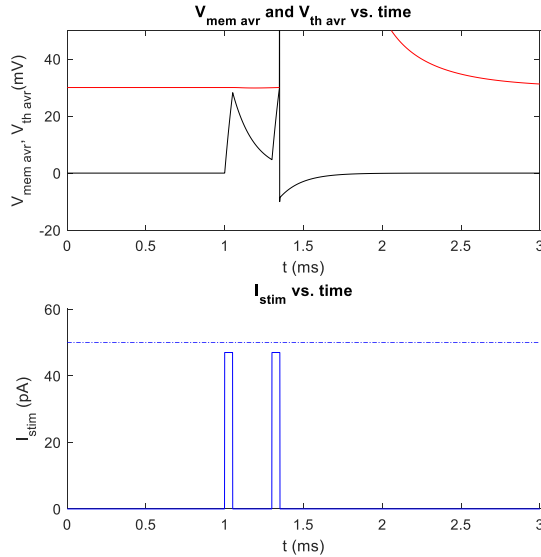


Figure 3.3. Single neuron membrane voltage response and firing threshold versus time for two closely spaced stimuli obtained using our model, showing the modeled effect of facilitation. All other neural phenomena, including noisiness, were disabled when producing this plot, and no spiking response was observed when active facilitation effect was deactivated, hence this is not a case of passive facilitation.

Upper plot: black: neural membrane voltage, red: threshold voltage; Lower plot: blue: applied stimulus, dashed black: threshold stimulus for causing a spiking response

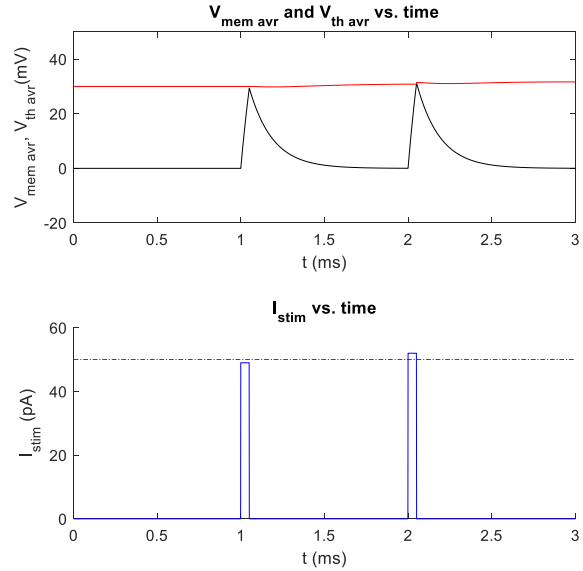


Figure 3.4. Single neuron membrane voltage response and firing threshold versus time for two closely spaced stimuli obtained using our model, showing the modeled effect of accommodation. All other neural phenomena, including noisiness, were disabled when producing this plot

Upper plot: black: neural membrane voltage, red: threshold voltage; Lower plot: blue: applied stimulus, dashed black: threshold stimulus for causing a spiking response

3.1.5. Spread of excitation

The spread of excitation in this model is considered in two separate parts: firstly, by considering the amount of the extracellular stimulus which reaches the auditory neuron fibers in general (i.e. the extracellular to intracellular spread), and secondly, by considering the variation of the intracellular stimulus which reaches different fibers along the electrode array (i.e. the longitudinal spread).

The extracellular to intracellular scaling is determined as in [Tabibi 2018]: namely, by measuring the extracellular stimulus subjective threshold (of fixed pulse width) and comparing it to the intracellular threshold obtained using this identical neuronal model, the calculated ratio describes the fraction of extracellular stimulus which reaches the auditory neuron fibers i.e. the effective intracellular stimulus. For example, in [Tabibi 2018] for a 26 μs stimulus pulses an extracellular threshold of 0.90 mA was measured, while the model-obtained intracellular threshold was 91 pA, thus giving a scaling of $91/0.90 \text{ pA/mA} = 101 \text{ pA/mA}$.

The longitudinal spread, i.e. the spread of excitation across different ANF fibers, is typically modelled as a decaying function with a peak at the stimulus location. To enable easier comparison to various studies, in this model it is possible to select two different spread profiles. Firstly, the exponential-absolute spread profile which has a sharp peak is modelled [Hess 2017, Hamacher 2004], which has the form:

$$f_{spread}(x) = \exp\left(-\frac{|x - x_{stim}|}{d_{spread}}\right) \quad (20)$$

where x_{stim} is the location of the stimulus, d_{spread} is the stimulus spread width, and x is the location of the observed auditory neural fibers. Second, the rounded exponential spread profile with a rounded peak is

modelled [Patterson 1982], which has the form:

$$f_{spread} = (1 + pg) \exp(-pg) \quad (21)$$

where p is the (inverse) stimulus spread width, and g is the normalized (relative) distance with respect to the farthest electrode from the stimulation location (i.e. zero corresponds to stimulus location, and maximum normalized distance is equal to one). Since the measured stimulus spread profiles are often asymmetric due to inherent asymmetry on the shape of the cochlea, the value of parameter p for each side of the cochlea can be taken to be different.

3.2. Model simulation

To solve all differential equations that occur in our model the Euler forwards method was used, though as shown in the results section, the midpoint method can offer either higher precision at identical timestep resolution, or coarser timestep resolution (i.e. faster runtime) at same precision. Since a timestep of 1 μs provides a satisfactory level of convergence, it was used to obtain all of the results shown below. The number of simulated neurons was initially kept low to enable faster execution and thus faster testing of influences of various parameters; by varying the number of neurons and observing the changes in response, it was determined that 1000 neurons are enough to arrive at a response similar to the response of a more realistic population of 10000 neurons [Tabibi et al 2018], which was mainly avoided due to impossibility of running the simulation. This has in turn prompted few attempts of code optimization which are outlined below.

In all cases, the applied stimulus was chosen corresponding to the available data: in general this implied a pulse width of 57 μs , inter pulse interval of 1 ms, and amplitude modulated stimuli, with square, sinusoidal and (reversed) saw modulation types and various modulation depths and frequencies. Note that our model is limited to monopolar stimulation and that only the cathodic phase of the stimulus is considered. Values of all neural parameters used in this model are shown in (Table 1) below.

Table 1.

Parameter	Code notation	Value
Membrane		
Resistance	R	1.9535 GΩ
Capacitance	C	0.0714 pF
Time constant	-	0.1395 ms
Resting voltage	-	0 mV
Threshold voltage	vth0	30 mV
Spike voltage	vspike	100 mV
Reset voltage	vreset	-10 mV
Threshold noise relative spread	RS	0.0774
Threshold voltage relative spread	Vth_stoch_norm	0.3
Refractoriness		
Absolute refractory period	T_ARP	0.332 ms (0.252 ms)
Relative refractory period	tau_RRP	0.411 ms (0.574 ms)
Relative spread strength	a_RS_ref	1
Relative spread time constant	tau_R_ref	0.2 ms
ARP/RRP relative spread	sigma_ref	0.1
ARP/RRP noise correlation	rho_corr	0.75
Spike rate adaptation		
SRA threshold strength	p_th_SRA	0.04
SRA threshold time constant	tau_th_SRA	50 ms
Relative spread strength	p_RS_SRA	0.04
Relative spread time constant	tau_RS_SRA	50 ms
Active facilitation		
Threshold strength	a_th_fac	-0.15 (ms) ⁻¹
Threshold time constant	tau_th_fac	0.5 ms
Relative spread strength	a_RS_fac	0.75 (ms) ⁻¹
Relative spread time constant	tau_RS_fac	0.3 ms
Accommodation (quick)		
Threshold strength	a_th_acc	0.5 (ms) ⁻¹
Threshold time constant	tau_th_acc	1.5 ms
Relative spread strength	a_RS_acc	0.75 (ms) ⁻¹
Relative spread time constant	tau_RS_acc	0.3 ms
Accommodation (slow)		
Threshold strength	a_th_acc_slow	0.01 (ms) ⁻¹
Threshold time constant	tau_th_acc_slow	50 ms
Relative spread strength	a_RS_acc_slow	0 (ms) ⁻¹
Relative spread time constant	tau_RS_acc_slow	50 ms
Spread of excitation		
Extracellular to intracellular spread scaling	scale_ex_to_in	91 pA / 0.9 mA
Length of neuronal array	d_max	2.25 mm
Location of stimulus	d_shift	0 mm
Spread width (exponential-absolute)	d_spread	0.8 mm
Spread width (rounded exponential, to base)	p_left	(4.1)
Spread width (rounded exponential, to apex)	p_right	(4.54)

Note: values in parentheses have been obtained by fitting to the ECAP recovery function and ECAP spread of excitation function (discussed in the next section) and were used to obtain all of the results in this study.

3.2.1. Code optimization

All computations referred in this section were performed on a dedicated desktop computer from Laboratory for Experimental Audiology, University Hospital Zurich, which has a processor i7 with 3.4 GHz maximum clock, 4 physical and 8 virtual cores, 8 GB of physical working memory (RAM, of unknown type) and 16 GB of virtual working memory.

During the initial phases of code development it was helpful to save the values of voltage, voltage threshold and x-factors describing each of the neural phenomena, and thus enable easier comparison of implemented code to the previous studies. However, using such code implementation, only a timestep resolution of 1 μ s for 60 ms stimulus duration and number of neurons of 1000 could be reached before the limit of computational power of used desktop computer was reached. Upon closer inspection in MATLAB and monitoring the system resource usage (e.g. in Windows Task Manager), the execution bottleneck was identified to come from a lack of RAM memory due to a large number of variables with dimensions ($num_iterations \times num_neurons$).

Since only the spike-times are important for predicting the ECAP response, in the “light” version of the implemented code only the instantaneous values of these variables are saved. Moreover, all variables data types have been recast from “double” to “single”, since in our case this does not reduce the precision of the model significantly and yet reduces the memory requirements. This approach has enabled simulating 10000 neurons while using 1 μ s resolution for 20 ms long stimulus duration, with a runtime of only 50 seconds, but simulations for longer stimulus durations proved to be too memory exhaustive.

The remaining memory bottlenecks were two variables of dimensions ($num_iteration \times num_neurons$): the applied stimulus and the voltage threshold noise. To reduce the memory usage for applied stimulus variable, a sparse coding approach was taken: since the inter-pulse interval is generally much larger than the pulse width, it is known that the applied stimulus mostly has a value of zero, hence only the nonzero values are saved. This is achieved by decomposing the applied stimulus variable into three distinct variables: binary variable of length ($num_iterations \times 1$) which shows whether a stimulus is applied at a given timestep or not, a variable which stores the nonzero values of the applied stimulus (length equal to ($num_pulses \times pulse_length$) where length is expressed in number of iterations), and a variable of length ($1 \times num_neurons$) which stores the spread of excitation for the given stimulus. To implement this in the code, at each timestep it is checked if a stimulus is applied, and if yes, the value of the compressed stimulus variables is accessed and then scaled according to the spread of excitation variable.

Now the voltage threshold noise remains as a bottleneck, and a straightforward solution is not possible in this case without slightly changing the systems dynamics or the spectrum of the noise (which has to be kept not too high). As an initial test, the noise was not precalculated as a ($num_iterations \times num_neurons$) long variable as before, but calculated anew at each N^{th} iteration as a ($N \times num_neurons$) long variable, thus reducing the instantaneous memory requirements. This approach has however produced a model which systematically predicted lower number of spikes, even when $N = num_iterations$ (i.e. the original method of noise calculation), and to date no clear cause of such behavior was found. Therefore, generating voltage threshold noise without strongly modifying its spectrum (and therefore spike times) remains an open question, whose solution should enable a noticeable decrease in memory requirements.

Looking further on, once the last remaining memory bottleneck is addressed, computational speed could be increased by executing the code on GPU instead of CPU. Namely, this approach would make use of many computational cores found in GPUs, and the fact that the implemented code is massively parallelizable since each neuron’s response can be simulated independently of the other neurons. The speed-up caused by parallel processing is already partly used, since use of implicit looping over neurons enables MATLAB® to perform parallel processing on the CPU, but performing parallelization on GPU promises a much higher impact on the computation runtime and should therefore be exploited in future work.

3.3. Neural Response Telemetry

We have now introduced all of the neural phenomena included in our model and their implementation, but how do we obtain the parameters necessary to describe these phenomena in CI patients? On the one hand, values of parameters determined in previous studies could be used, while on the other hand one could also obtain these parameters from suitable measurements. In this study, the first approach is used to initialize our parameters, with values being taken mostly from [Boulet et al 2018] for temporal phenomena and [Hamacher 2004] for ARP/RRP stochastics and stimulus spread, while the other approach is used to more precisely fit the parameters to each individual patient, and thus produce better predictions.

When performing animal studies, it is possible to invasively measure the response of single neurons or small patches of neuronal fibers [Miller et al 1999] and thus obtain the necessary parameters, but in clinical applications such approach is impossible. However, the same electrode array used for stimulating the auditory neuron fibers can be also used to record the voltage sensed by the electrodes, thus providing an objective measure of neuronal response [Dillier et al 2002] (Fig 3.5.).

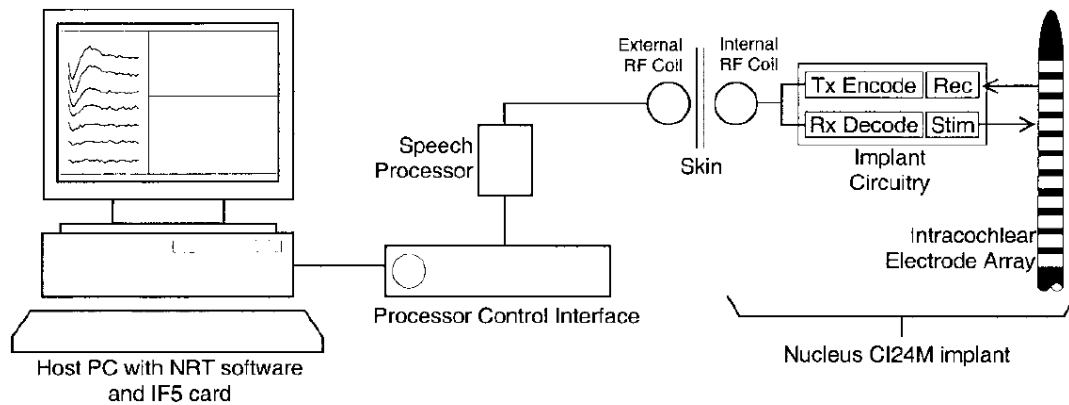


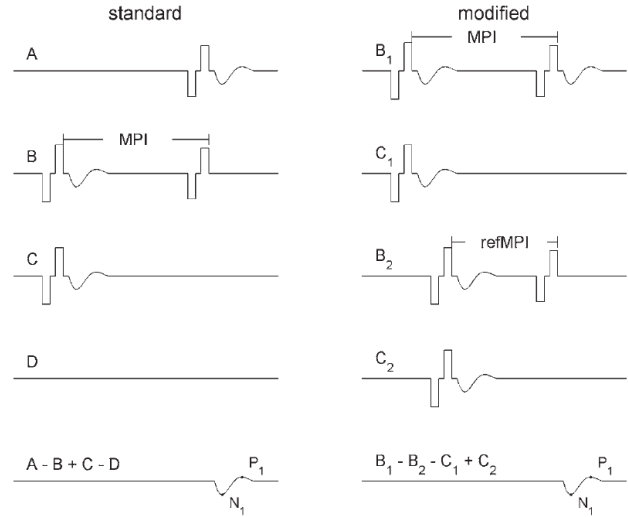
Figure 3.5. Neural response telemetry setup, showing the typically measured evoked compound auditory potentials (ECAPs) on the left. Source: [Dillier et al 2002].

The measured signal is, though, not a local measure of neural activity, but a sum of all action potentials (spikes) produced by all neurons in auditory neuron fibers – hence the name *evoked* compound action potential. Due to differing latencies, i.e. times at which individual spike are generated, this signal is also not a simple multiple of the number of spiking neurons and the amplitude of a unit action potential, but is slightly lower depending on the amount of synchronicity in the neural response (which tends to increase as stimulus level is increased). Likewise, the measured signal depends on the number of surviving neurons and their excitability at the time of stimulus application.

To obtain the ECAP signal, due to the artefact caused by applying the probe stimulus, a subtraction paradigm developed by [Abbas et al 1990] and modified by [Miller et al 2000] has to be used. To cancel out this probe artefact, a measurement with masker and probe (labelled B in figure 3.6.) is performed, where the masker completely masks the probe (i.e. masker-probe interval is smaller than the absolute refractory period), thus leaving only the probe artefact to be measured. However, this measurement now contains the masker response and masker artefact as well, which need to be cancelled by an additional masker only measurement (C). Finally, the measurement noise is measured (D), and the measurements are combined, leaving only the probe response shown in bottom left in (Fig. 3.6.).

When aiming to obtain an ECAP response for (modulated) pulse trains, this method is only slightly adapted: namely, when calculating the response to the n^{th} pulse in the pulse train, all of the preceding $n-1$ pulses are regarded as masker pulses. Subtracting this signal produces the desired neural response to the n^{th} pulse and the probe artefact, which can be then cancelled in the same fashion as in a typical measurement method.

Figure 3.6. Left: standard NRT artefact cancellation technique [Abbas et al 1990]. Right: modified NRT artefact cancellation technique by [Miller et al 2000]. Source: [Morsnowski et al 2006]



3.3.1. Amplitude growth function

After obtaining a single ECAP response, a logical next step is to determine how does the ECAP amplitude (defined as the difference between the 1st negative and 2nd positive peak in the response) change with the strength of the applied stimulus, where the stimulus is a single pulse. This relation is referred to as the amplitude growth function (AGF) and is measured by first finding the loudest acceptable stimulus level for a given patient, and then recording the ECAP amplitudes as the stimulus strength (of a single pulse) is reduced, until no response can be measured. If thus obtained AGF relationship is plotted using the SI units for the applied stimulus (typically μA), one would obtain a logarithmic curve. Due to practical reasons, a unit called Current Level (CL) is clinically used where the relation between stimulus measured in CL units and SI units is given by:

$$I[\mu\text{A}] = 17.5 \mu\text{A} * 100^{CL/225} \quad (22)$$

The resulting curve is shown in the (Fig. 3.7.) and as desired, the relation between ECAP amplitude and CL is linear. From the practical point of view, this enables an easier prediction of ECAP amplitudes one could observe: e.g. for a stimulus which is (amplitude) modulated in CL units, due to the linear relation one would expect to observe a similar amplitude modulated ECAP response, with the modulation depth depending on the slope of AGF function. However, one must be aware that AGF is measured using single pulses, and thus it is not sensitive to any of the temporal neural phenomena described earlier, producing a noticeable difference between the expected and observed responses [Jeng, Abbas et al 2009].

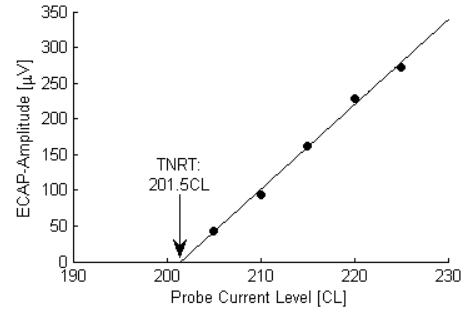


Figure 3.7. Example of an amplitude growth function, with the extrapolated NRT threshold shown. Adapted from [Hess 2017].

Another important fact to note is that the extrapolated ECAP (or T-NRT) threshold is higher than the subjective threshold i.e. with an increasing stimulus the patient first hears a sound and only later an ECAP response can be noticed. This happens due to a lack of synchronicity in neural response at low stimulus levels, making the individual responses only partly overlap, and thus producing a weak signal which can fall below the measurement noise level. In some cases reported by [Dillier et al 2002], this can lead to no ECAP response being measurable with some patients.

However, in most cases an ECAP response can be observed, which enables using the measured AGF to extract some of the relevant neural parameters. Tentatively, the maximum ECAP amplitude is achieved for high stimulus strength and thus high neural synchronicity, and in the first approximation of no latency, it can be assumed that the ECAP amplitude is proportional to the total number of spiking

neurons for the applied stimulus. Due to the spread of excitation, this is yet again not a local measurement, but one including a weighted average of all neurons in the population, where the weighting factor is given by the spread of excitation function and the local threshold (distribution) function. Given measurements across multiple electrodes of both AGF and the spread of excitation, one could then combine these measurements into a matrix and invert it, thereby obtaining a total number of surviving neurons at each electrode. This however depends on many assumptions, and thus as a rough estimate one could consider the maximum ECAP amplitude to be informative of the local number of surviving neurons, and only then attempt to derive a more precise measure.

3.3.2. Recovery function

A method used to characterize refractoriness of a single neurons was to use a suprathreshold masker and a probe, and observe for which probe levels one would obtain a spiking response. An identical method can be applied for a neural population using ECAP measurements, i.e. using a probe to measure the remaining excitability of a neuronal population after a spiking response to the masker pulse. For short masker-probe intervals one would expect no ECAP response, while for longer intervals one would expect the response to slowly return (recovery) towards the resting response. Plotting the ratio of measured to rest ECAP response for various masker-probe intervals, one obtains a curve (called recovery function) as shown in the (Fig. 3.8.) below.

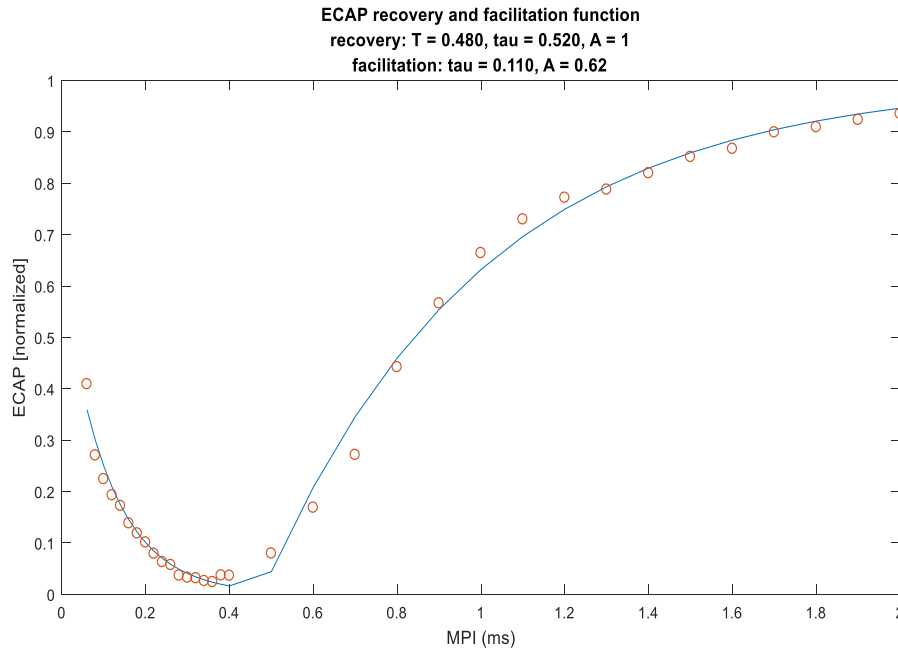


Figure 3.8. Model-obtained recovery and facilitation function, shown together with fitted functions. Number of neurons was set to 1000, duration of ARP and time constant of RRP were set to 0.5 ms, and masker and probe pulse had an equal level corresponding to twice the threshold level (in mA).

Red: model obtained ECAP values for various masker-probe intervals. Blue: fitted curve, which was assumed to be a combination of an exponentially decaying term (modelling facilitation) and unity minus decaying exponential term delayed by a certain offset, modelling refractory recovery. Fitted parameters are shown above the figure, with values matching the user predefined parameter values, demonstrating the ability of such method of patient parameter fitting.

Interestingly, the region of short masker-probe intervals shows an increase as the interval is reduced, i.e. a facilitation-like behavior. The reason such a behavior is observed is again due to the population effect: alongside the spiking neurons which are situated close to the active electrode, there are neurons further away which have not spiked but have reached a voltage close to their (rest) threshold, and given a stimulus at short interval, these neurons can indeed spike – a clear case of facilitation. That this is indeed the case can be vividly seen in (Fig. 3.9.), which shows the indices of firing neurons versus time: as expected, it is the peripheral subpopulation of neurons which produces the facilitatory response.

Given all of the above, when dealing with a neural population, one will record both the recovery and the facilitation function at the same time. This in turn enables obtaining ARP/RRP parameters (their time scales) and facilitation parameters, with an important note that both passive and active facilitation are measured as one combined phenomena, and their effects cannot be disentangled at this point. Performing such measurements at different electrodes can potentially provide a measure of variations in ARP/RRP parameters, which has been observed in auditory neuron fibers [Morsnowski et al 2006].

Another important point is that the measured recovery and facilitation function have been observed to change with the stimulus level, with the time scale of recovery (the RRP constant) becoming longer and the extrapolated maximum of facilitation function (at zero masker-probe interval) becoming larger for stronger stimulus levels. Here these variations have not been considered, as they would have required analyzing the obtained parameters for various stimulus levels.

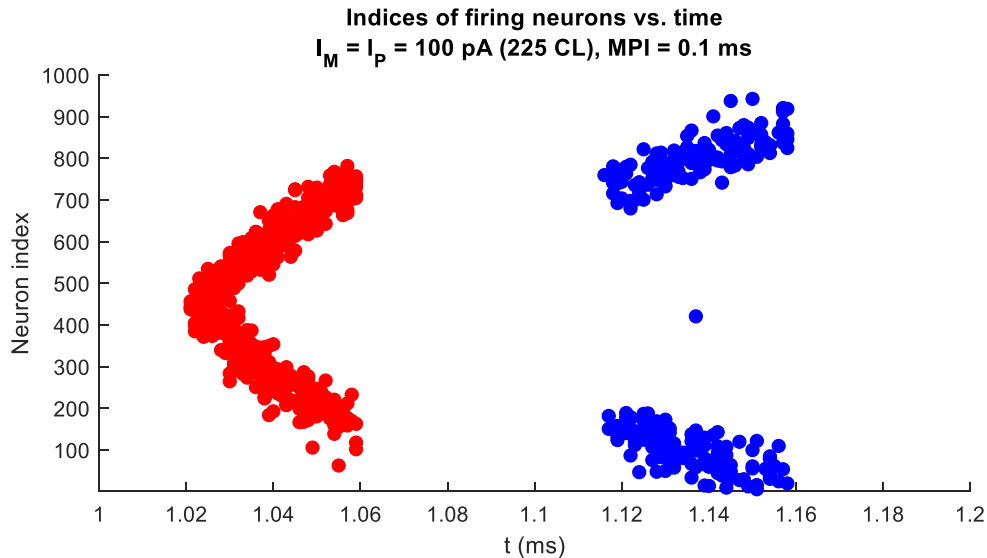


Figure 3.9. Indices of firing neurons versus time for suprathreshold double pulse stimulus, with masker-probe interval falling inside the facilitation time scale. Stimulus was applied at the position of 450th neuron. As expected, neurons firing to the second pulse are the peripheral neurons (blue), as the neurons closer to the stimulating electrode are still in the absolute refractory period (red). Overall of firing populations is primarily a visual effect due to the size of markers used, as mean duration of absolute refractory period is longer than the considered masker-pulse interval.

3.3.3. Spread of excitation function

As was hinted in the previous section, the spread of excitation can be determined by using the ECAP measurements. A wrong method to do so would be to apply a stimulus at a certain location and record the ECAP amplitude at each electrode in the array, the reason being that this measures the spread of electrical fields generated by neuronal action potential [Cohen et al 2003].

To measure the spread of excitation, it is necessary to record the ECAP response to a probe applied at a certain fixed location, while a masker pulse is applied at various location across the electrode array [Cohen et al 2003], where the MPI interval is kept smaller than the ARP duration in order to achieve maximum masking. In such a way, one measures the remaining excitability at the probe location, which depends on the value of the electrical field the masker pulse produced at this location, but also the (local) neural density. By taking the value of the ECAP response when a masker is applied far away and subtracting the measured ECAP response from this reference, a spread of excitation function at various electrode locations can be obtained (typically the obtained spread functions are not symmetrical, and thus separate spread width can be used for each side of the function). This information enables one to, for example, decide for which electrodes it is necessary to use more focused stimulation (e.g. bipolar stimulation) or which neurons are likely to be unexcitable following an excitation at a nearby electrode, as performed in the RSC coding strategy [Babacan 2010].

3.3.4. ECAP model

As already implicitly suggested in previous sections, the data measurable from CI patients consists of ECAP measurements (also see below), but our model however does not produce the predicted ECAP response, but the (histogram) of spike timings. To enable comparison of our model predictions with data, it is thus needed to develop an additional model which calculates the “synthetic” ECAP response from a given spike timing history.

This can be achieved in the manner described in [Hamacher 2004 and Hess 2017]: for each stimulus pulse, at each spike time a unit action potential is placed, and from the resulting compound action potential only the amplitude (N1 to P2 peak) is saved as the ECAP amplitude at that stimulus pulse. Mathematically this corresponds to a convolution of the spiking histogram with a unit action potential, and here the shape (temporal widths and amplitudes of N1 and P2 peaks) of the unit action potential has been taken to be identical to the one defined in [Hamacher 2004 and Hess 2017].

3.4. Data

Data used for obtaining results in this study was collected during a previous project by [Hess 2017], where using NRT measurements, ECAP responses for a single patient were collected for amplitude modulated pulse trains of various type and frequencies, with inter-pulse interval of 1 ms, pulse width of 57 μ s and for a total duration of stimulus of 60 ms. Modulation types were “square”, “sinusoidal”, “saw” and “reverse saw”, maximum CL level was set equal to the loudest acceptable CL, modulation frequency was set to either 50 Hz or 100 Hz, and the modulation depth was between 2% and 5% in (relative) CL units. Additionally, values of ARP duration, RRP time scale and facilitation time scale were obtained from recovery and facilitation functions measured in another study currently underway [Tabibi et al 2018].

3.5. Parameter optimization

Owing to the variability of many neural parameters between patients (e.g. caused by differences in the stage of neural degeneration), it is generally considered that making neural parameters individualized for each patient will produce better model predictions. This approach is only partly followed in this study though, due to a lack of data covering the whole spectrum of measurements necessary to individualize our model parameters, and only the ARP/RRP durations and spread widths are individualized, while other parameters are taken from [Boulet et al 2018] study, which relies mostly on feline auditory neuron data.

Since optimizing all of the neural parameters at the same instant would be ill-advised due to an overwhelming parameter space (as there are over 30 parameters), a better method would be to optimize only small subsets of parameters at one instant. For example, when considering the measured recovery and facilitation function (at one electrode), one would optimize only the values of ARP/RRP duration, membrane time constant, and active facilitation strength and time constant, and then go on to optimize other parameters.

If such parameter individualization were to be performed (as partly discussed in section 3.3.), one can envision the battery of measurements to be performed as described in Table 2. Note that whenever possible, measurements should be performed at as many electrodes as possible in order to resolve the local differences in neural parameters, but when this is not the case, one can interpolate the values of parameters obtained for e.g. 3 electrodes placed at the base, center and apex of the electrode array.

After determining as many neural parameters possible beforehand, one can perform optimization by comparing predicted and measured ECAP responses to (un)modulated pulse train stimuli, where the cost functions (i.e. metrics of mismatch) could be the squared differences between measured and predicted signals, comparisons of maximum achieved ECAP amplitudes and their changes from one modulation period to the other, amount of jitter in ECAP responses etc. The algorithms to perform optimization could range from quite simple ones, such as specifying the range of parameter values and finding a value with the least squared error (implemented in the graphical user interface shown below in

Fig. 3.10.), towards more complex algorithms which stochastically sample the parameter space (thus avoiding the “curse of dimensionality” i.e. long runtimes), such as particle swarm optimization and similar. To reduce the runtime of such advanced algorithms, however, one would need to resolve the computational bottlenecks discussed previously, especially since the optimization procedure for numerous parameters would be suitable for GPU computations. In addition, note that all neural phenomena could be in principle obtained without resorting to measurements with modulated pulse trains, however these measurements still need to be performed since, on the one hand, this enables one to validate the values of parameters obtained with preceding measurements, and on the other hand, enables one to optimize the neural parameters against a more realistic stimulation scenario, and resolve differences between possibly multiple sets of parameters which yield identical errors in optimization algorithms when considering only the prior measurements.

Table 2.

Measurement	Measurement parameter	Obtainable neural parameter
Clinical map	Comfort and threshold levels	Extra- to intracellular spread scaling Threshold voltage (first approximation using previous values of membrane capacitance)
Amplitude growth function	General Maximum ECAP amplitude and AGF slope ECAP threshold (T-NRT)	Expected modulation depth in ECAP response to modulated pulse trains Locally weighted number of surviving neurons Number of neurons needed to produce a measurable ECAP signal
Spread of excitation	General Combined with AGF and measured at various electrodes above + SoE measured at various stimulus levels	Spread width parameter(s) (unless the measured SoE function is loaded “directly” into the model) Local number of surviving neurons (using matrix inversion, as described in text) Distribution of threshold voltages (tentative, using matrix inversion).
Recovery function / facilitation function	General Measured at various electrodes	ARP/RRP duration, passive and active facilitation time scales, active facilitation strength ARP/RRP duration relative spread and correlation
Unmodulated pulse train ECAP responses	General (possibly for various inter-pulse intervals)	Spike rate adaptation and accommodation strengths and time scales, possibly modifications to other parameters
Modulated pulse train ECAP responses	General (with various stimulus parameters)	As above

3.6. Graphical user interface

To enable the future users of this code to easily change any neural parameters and test various model variants (i.e. combinations of neural phenomena), a graphical user interface (GUI) was developed using the MATLAB® GUIDE development tool, shown below in (Fig 3.10.).

Besides the ability to arbitrarily combine neural phenomena (including spread of excitation), it is possible to fix the generated noise variables, thus enabling easier comparison of results of various model variants. Additionally, one can vary the simulation parameters: number of simulated neurons, timestep resolution, duration of simulated “experiment”, and whether one would like to use the “light” version of the code (where membrane voltage and threshold voltage are not saved), or the “heavy” version of the code which retains all of the model data during each iteration.

To facilitate comparison to various studies and testing the neural responses to single and double stimuli, the applied stimulus can be chosen to be either single, double or (un)modulated pulse train, with modulation being performed in CL or SI units. Naturally, stimulus strength, pulse duration, inter-pulse intervals, modulation type, frequency and depth can be all changed at will.

Spread of excitation parameters can be also changed, together with the stimulus location, length of electrode (i.e. neuronal) array, and additionally, the type of spread excitation can be changed between exponential-absolute [Hamacher 2004] or rounded-exponential [Patterson et al. 1982], where in the latter case the spread parameters can be made asymmetric by the user. Currently the stimulus location is fixed during one simulation run, but in future versions this could be changed to enable predictions to more realistic simulation scenarios.

To enable comparison of model-predicted vs. measured ECAP responses, the GUI enables loading and plotting the loaded ECAP data, and one can choose if the data and the model-predicted ECAP responses will be normalized or not. Additionally, the maximum applied stimulus level, modulation depth and modulation frequency are automatically extracted from the loaded data and shown, though this feature is currently possible only for the data with format identical to the data from [Hess 2017].

Building on top of the ECAP response model, synthetic measurement of the amplitude growth function and recovery functions can be automatically performed, which enables comparison of the model predicted AGF and recovery function to the expected ones. Currently it is not possible to load the measured AGF and recovery functions, though this feature can be later implemented relatively easily, thus enabling one to automate fitting of parameters extractable from AGF, and recovery and facilitation functions.

Each of the generated figures can be independently saved and edited, thus enabling faster generation of figures needed for various reports etc, and in the bottom right a simplistic optimization panel is included. Here one can choose an arbitrary variable, and by setting the lower and upper bound and the step size enable the optimization algorithm to calculate the squared error between predicted and data ECAP response for each value of the parameter, and thus output the optimal value at the end. When necessary, the user can export all of the model calculated data to a .mat file and perform further work “offline”, or return to the initial settings by resetting all parameter values using a single button.

CHECK

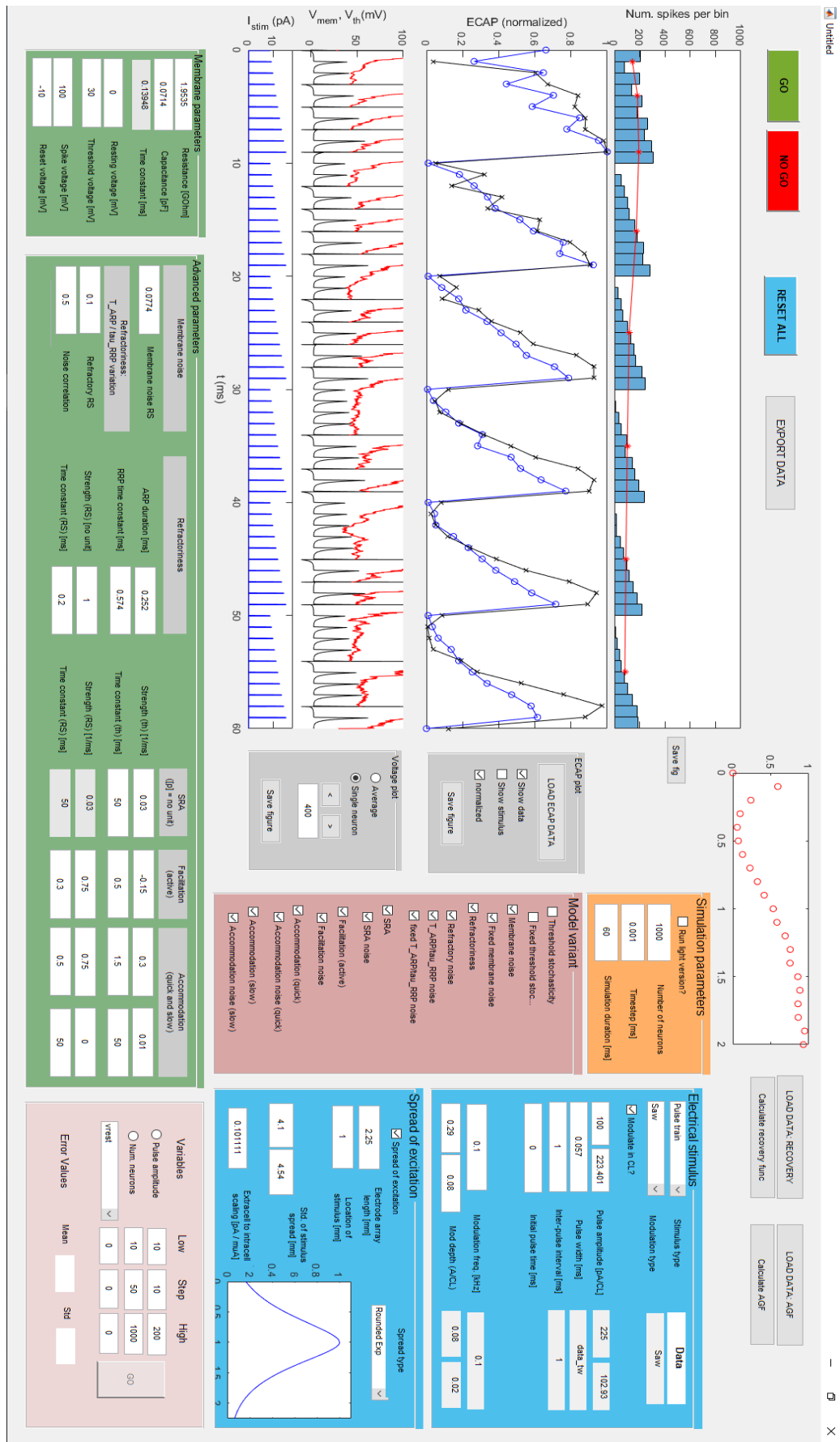


Figure 3.10. Screenshot of the implemented GUI, showing modelled and data ECAP response, predicted spike histogram, stimulus strength, voltage plot of a single neuron, spread of excitation and various panels in which neural, simulation and stimulus parameters can be adjusted by will.

4. Results and Discussion

4.1. Simulation timestep convergence

To ensure that convergence of results has been achieved with respect to the time step resolution, the model current threshold was determined for various time step resolutions and for two different time stepping methods. As shown in the Fig 4.1. below, for a pulse width of 50 μs and with threshold noise deactivated (to enable easier comparison), the value of current threshold grows with increasing temporal resolution, with convergence being achieved only slightly below the resolution of 1 μs . The primary reference of this study [Boulet et al 2018] consistently used a temporal resolution of 1 μs and Euler forwards method, which produces an acceptable error of 2% in the firing threshold, while the secondary reference [Hess 2017] used a timestep of 10 μs and Euler forwards method (see GUI function of this reference), which produces a large 14% error in the firing threshold, and thus in all results produced in that study.

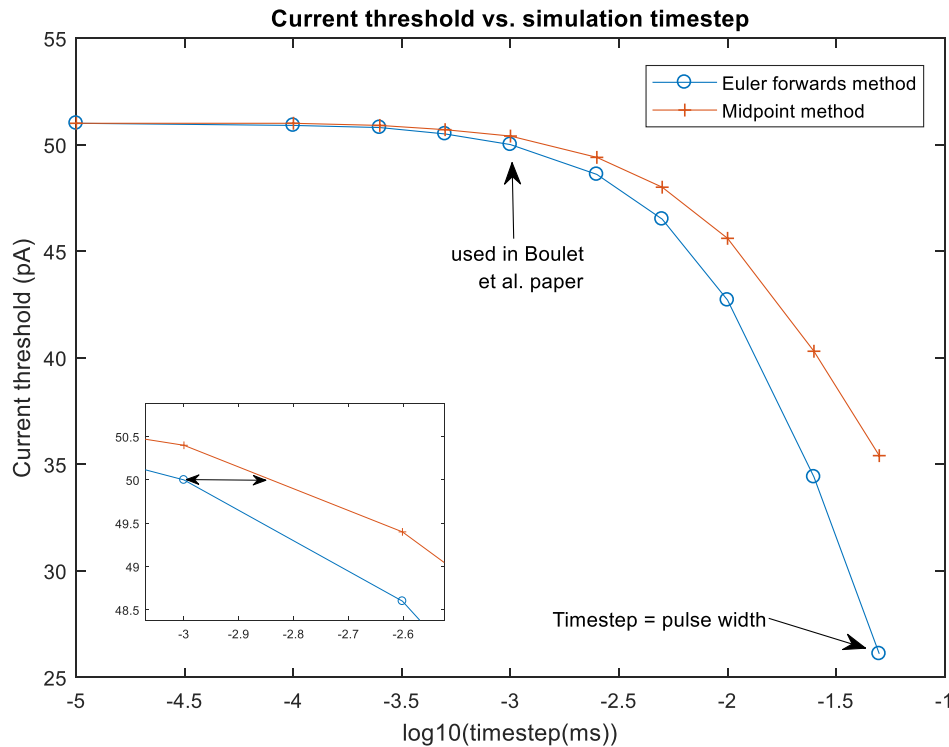


Figure 4.1. Dependence of single fiber current threshold versus time step resolution for Euler forwards and midpoint methods. Results have been obtained using stimulus with pulse width of 50 μs and with threshold noise being deactivated to enable easier and faster comparison. Arrows show the time step resolutions corresponding to timestep equal to pulse width, producing an error of 50%, and to time step of 1 μs used in [Boulet et al 2018].

Inset: zoomed in region around time step of 1 μs , with arrow showing the equivalent time step with midpoint method which would produce results with identical precision.

The decrease in the firing threshold with coarse time step resolution can be attributed to the following fact: when the membrane voltage is increasing during a stimulus, between two timesteps the discharge term ($-v/\tau$) has an average value which is higher than its value at the beginning of the timestep. Therefore, the modeled discharge term is smaller than it is in reality, making the achieved voltages higher in general (opposite is true when no stimulus is applied i.e. when the membrane voltage is decreasing). Similarly, when a time step coincides with the end of the stimulus pulse (or is near its end), with the Euler forwards method only the current value of the stimulus is sampled (which is nonzero) and the stimulus is assumed to remain constant until the next timestep, thus artificially prolonging the pulse stimulus and making it stronger, leading to lower firing thresholds.

Therefore, an attempt to decrease the simulation runtime by reducing the temporal resolution could produce erroneous results and must be done with proper care. A simple method to achieve this is by using a higher-order method for time stepping, as for example the midpoint method. In such case, the

same method has to be used in updating all other variables whose evolution is described by differential equations, such as voltage threshold and noise factors described earlier for all neural phenomena.

A demonstration of convergence achieved when using the (implicit) midpoint method is shown in the (Fig. 4.1.) above: convergence is generally faster, and compared to the timestep value used in [Boulet et al 2018], the same precision is achieved with 40% longer timestep (1.4 μ s versus 1 μ s, see inset), with only a minor code modification i.e. rewriting one line of the code as below:

$$\frac{dv}{dt} = \frac{v(i+1) - v(i)}{\Delta t} = -\frac{v(i) + v(i+1)}{2RC} + \frac{I(i) + I(i+1)}{2C} \quad (23)$$

$$v(i+1) \left(1 + \frac{\Delta t}{2RC}\right) = v(i) - \frac{\Delta t}{2RC} v(i) + \frac{\Delta t}{2C} (I(i) + I(i+1)) \quad (24)$$

4.2. Unmodulated pulse train

As single neuron responses to pulse train stimulation has been already presented in our main reference [Boulet et al 2018], and the population responses do not significantly differ from single neuron responses, here we shall only replicate one of the results from that study.

Shown below is the evolution of firing rate with time for various inter-pulse intervals (i.e. stimulus rates), with figures 4.2. and 4.3. showing results for a model with spike rate adaptation and accommodation (both quick and slow) disabled and enabled, respectively. In both cases refractoriness and threshold noise were enabled. As expected from previous studies, one can see the firing rates drop and even reach zero (no firing) for higher pulse rates, despite the general growth in firing rates one would expect when spike rate adaptation and accommodation are not present.

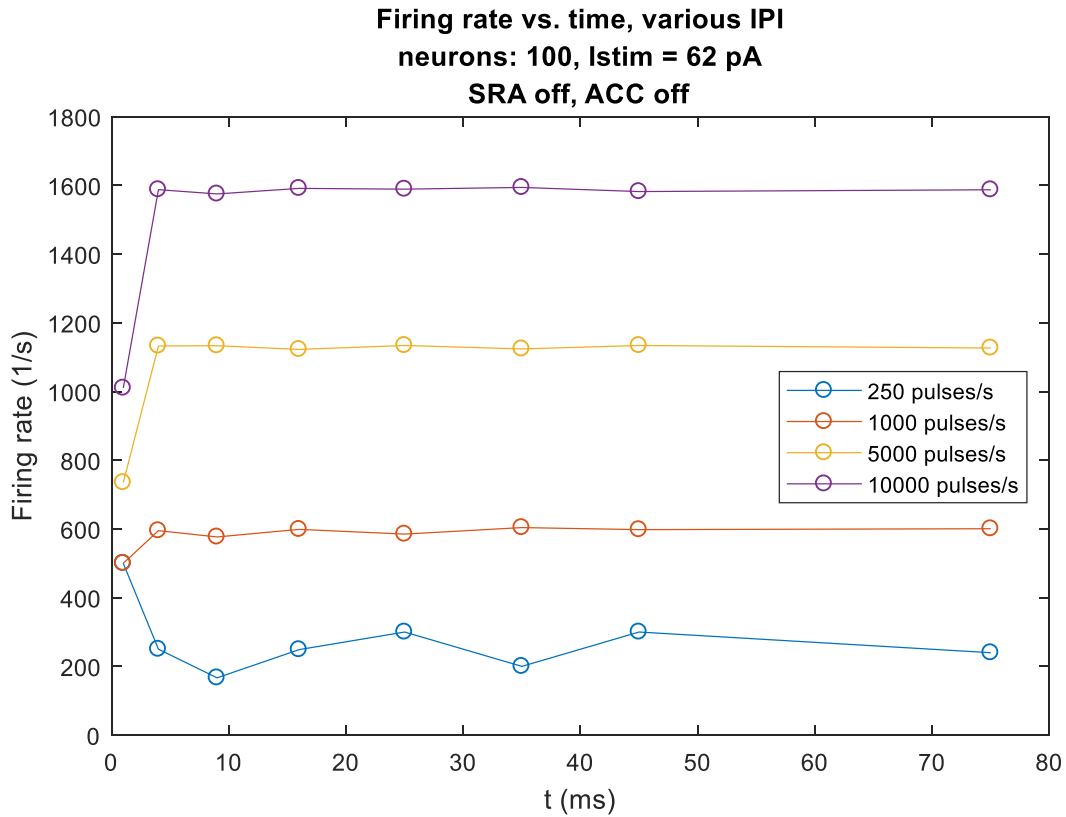


Figure 4.2. Firing rate versus time for pulse trains stimulation of various stimulation rates / inter pulse intervals, with the spike rate adaptation and accommodation effects disabled. Amplitude of pulse train was set to 62 pA, pulse width to 50 μ s, and number of simulated neurons was 100.

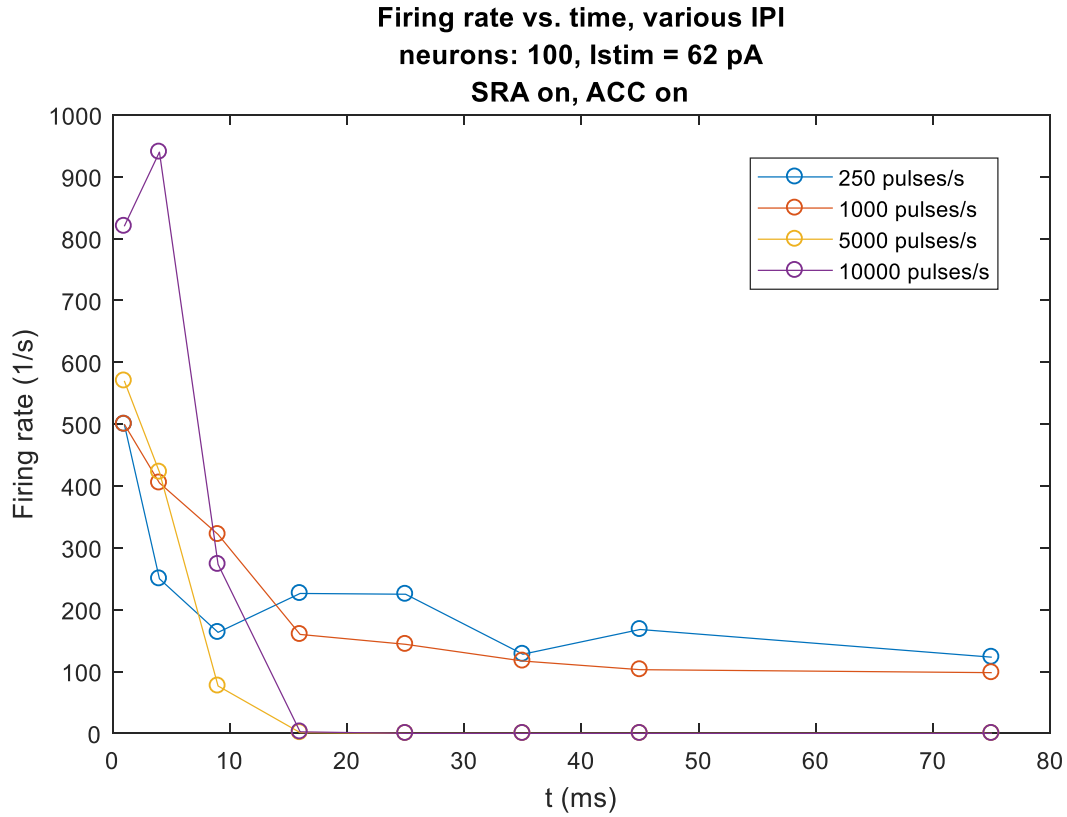


Figure 4.3. Firing rate versus time for pulse trains stimulation of various stimulation rates / inter pulse intervals, with the spike rate adaptation and accommodation effects enabled. Amplitude of pulse train was set to 62 pA, pulse width to 50 μ s, and number of simulated neurons was 100. A noticeable drop in firing rates compared to the previous case can be seen, and for two highest pulse rates the neurons stop firing all together.

4.3. Modulated pulse trains: square/step modulation

If one attempts **CHECK** to fit the model predictions to ECAP data for square modulated pulse trains, a distinct issue shown in Fig. 4.4. below will be noticeable: even though the general shape and the oscillating nature of response to a constant stimulus signal are apparent, the modulation depth of the predicted ECAP response seems to be much shallower than observed in data, i.e. the response to weaker parts of the stimulus is too strong.

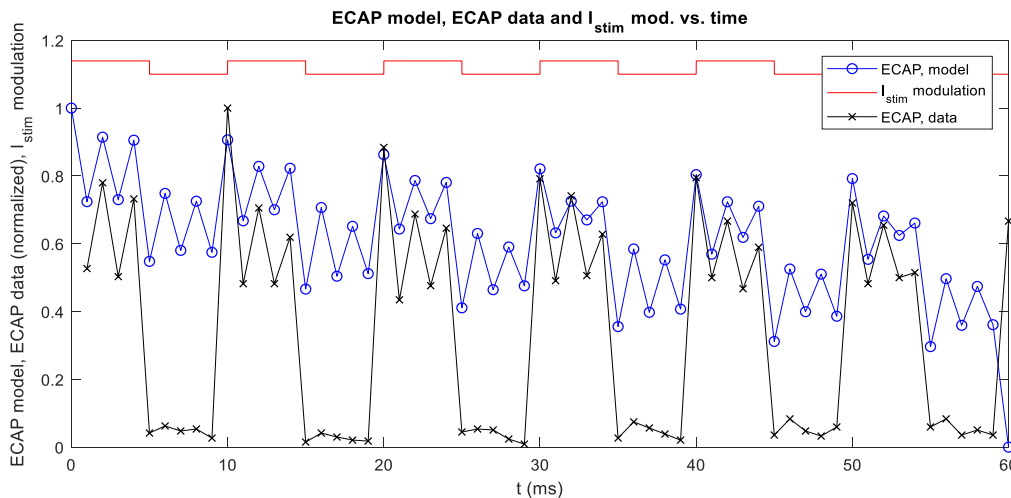


Figure 4.4. Predicted and observed normalized ECAP response for square modulated pulse trains, with modulation envelope shown above with arbitrary offset. Maximum stimulus level is 215 CL, modulation depth 2% (in CL), modulation frequency 100 Hz, IPI 1 ms, SRA strength reduced to 0.01, accommodation strength reduced to 0.3,, and all neural phenomena are enabled. Stimulus spread was of rounded-exponential type and stimulus was applied at the center of the neural population.

Tentatively, one can attempt to solve this by increasing the threshold or by reducing the amount of extracellular stimulus reaching the auditory neural fibers, thus forcing the response to weaker phases of the stimulus to fall below the T-NRT. Results of such modification are shown in Fig 4.5., and though helpful, the discrepancy in response modulation depth remains since the above modifications affect both the stronger and weaker phases of the stimulus pulse.

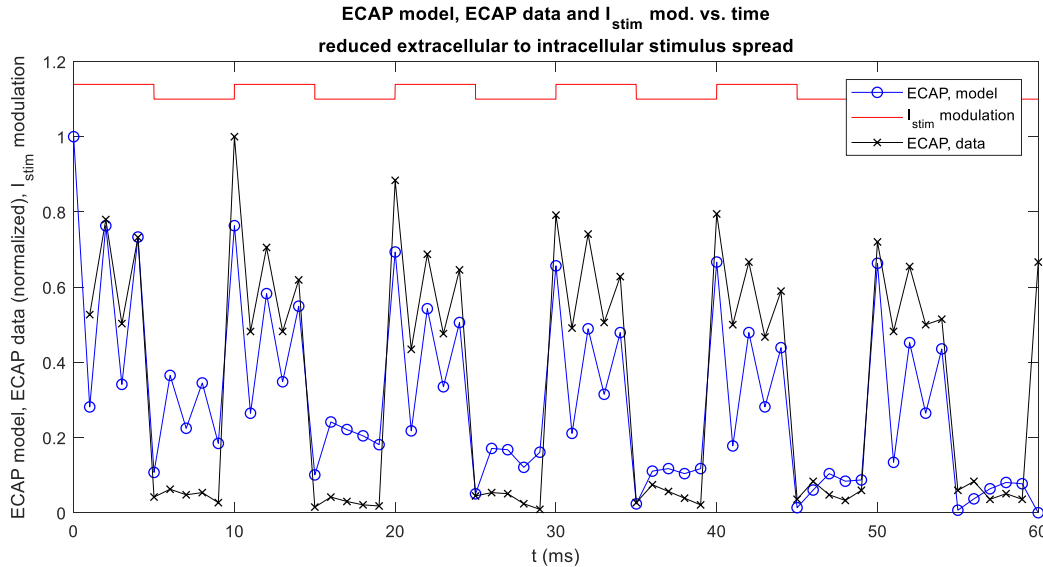


Figure 4.5. Predicted and observed normalized ECAP response for square modulated pulse trains, with modulation envelope shown above with arbitrary offset. Same parameters as in Fig. 4.4. except the extracellular to intracellular spread which was changed from 0.101 to 0.060.

The same discrepancy has been observed for all other modulated pulse trains (with various modulation depths and frequencies), suggesting that the underlying cause is of general nature. One possible reason for this can be understood by considering the synthetic amplitude growth function our ECAP model would produce: contrary to the behavior observed in patients, our model predicts a measurable ECAP response as soon as a single neuron has fired, i.e. at the single neuron firing threshold, thus making the subjective and objective (T-NRT) threshold equal. However, as mentioned earlier, in practice the objective threshold is found higher than the subjective one due to an insufficient amount of synchronicity in spiking of neural fibers (i.e. large spread in spike latencies). Since our ECAP model assumes no latency, thus partly matching the amplitude growth function for higher stimulus levels, increasing the model T-NRT would lead to an increase in slope of model AGF, thus leading to higher modulation depth in ECAP response. As this influences all model predictions, the first step of any further work should be to improve on our ECAP model, which is currently the “weakest link” in this study.

However, it is still possible to analyze the obtained data using our model: by increasing the stimulus modulation depth, an effect similar to the above can be achieved, though at the cost of having the modulation depths in model and data mismatch (note that this method was implicitly used in [Hess 2017], but with large differences in model vs. data modulation depths). Therefore, from here on all of the reported model predictions are obtained using an increased modulation depth relative to the data.

Results obtained in such a manner are shown in three panels of figure 4.6. below, where modulation depth was increased to 6% (3 times higher than in the data) and the SRA strength was varied to show its (strong) effect on the response and to find its optimal value. The general shape and response modulation depth can be fitted in a satisfactory way, though the weaker parts of the stimulus still produce a higher than observed response, suggesting need to further adjust (i.e. determine from subjective threshold) the extracellular to intracellular spread scaling. Additionally, the amplitude of oscillations in the response do not seem to weaken as much as suggested by the data: noise and ARP/RRP duration relative spread were modified in an attempt to fit this behavior, however only a minor improvement could be observed, thus suggesting further investigations are necessary. Note that in all cases showing comparisons of various model parameters, to enable easier comparison the calculated noise and ARP/RRP noise variables have been kept the same across all considered cases.

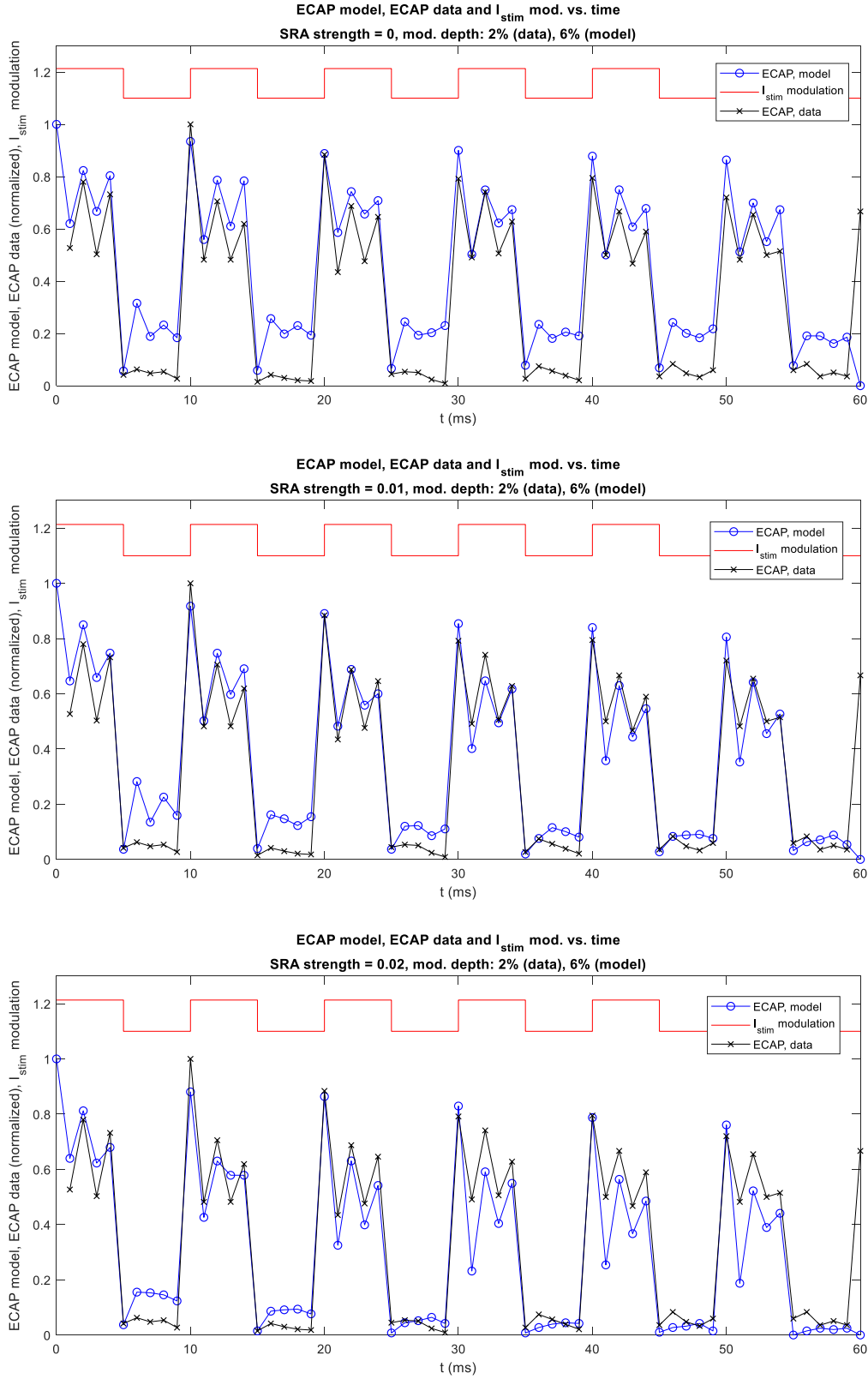


Figure 4.6. Predicted and observed normalized ECAP response for square modulated pulse trains for various SRA strengths, with modulation envelope shown above with arbitrary offset. Same parameters as in Fig. 4.5. except the extracellular to intracellular spread which was kept at 0.101, modulation depth was increased to 6% (3 times higher than in the data), and the SRA strength was varied from zero (top), to 0.01 (middle), and to 0.02 (bottom).

4.4. Modulated pulse trains: sinusoidal modulation

If we now shift our attention to the sinusoidally modulated pulse trains and use the same parameter set as in the middle panel of Fig. 4.6., the results such as shown below in Fig. 4.7. will be obtained. Compared to the stimulus, the data ECAP exhibits a slight negative phase shift i.e. it reaches its maximum before the stimulus, and a decrease/sharpening of the response with each ongoing pulse can be noticed. The phase shift can be explained from the fact that most of the neurons are excited already slightly below the maximum stimulus level, thus even though the stimulus is still slightly increasing the overall excitability will decrease more strongly, producing a lower ECAP response (mathematically one can approximate the sinusoidal function near the maximum with a constant stimulus, which we have seen to produce an oscillatory response i.e. decrease in excitability following the first pulse). As for the reduction in ECAP response width with time, this can be attributed to an on ongoing decrease in excitability of neurons, and as visible in further figures below, spike rate adaptation (and to some extent accommodation) are responsible for such behavior.

Given all of the above, the predicted ECAP response below suggests that either SRA or accommodation (or both) need to be increased to reach more satisfactory results. As an addition, below we provide a figure of predicted ECAP response for higher stimulus modulation depth of 10% (Fig. 4.8.), which does fit the response modulation depth observed in the data, but fails to produce the phenomena described above.

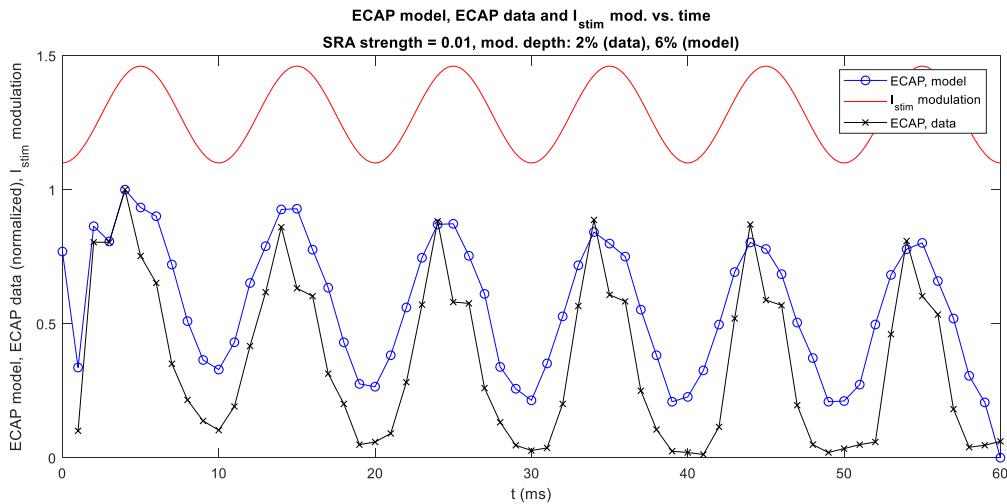


Figure 4.7. Predicted and observed normalized ECAP response for sinusoidally modulated pulse train, with modulation envelope shown above with arbitrary offset. Same parameters as in the middle panel of Fig. 4.6. except that the maximum stimulus level is 225 CL (i.e. higher as compared to previous figures).

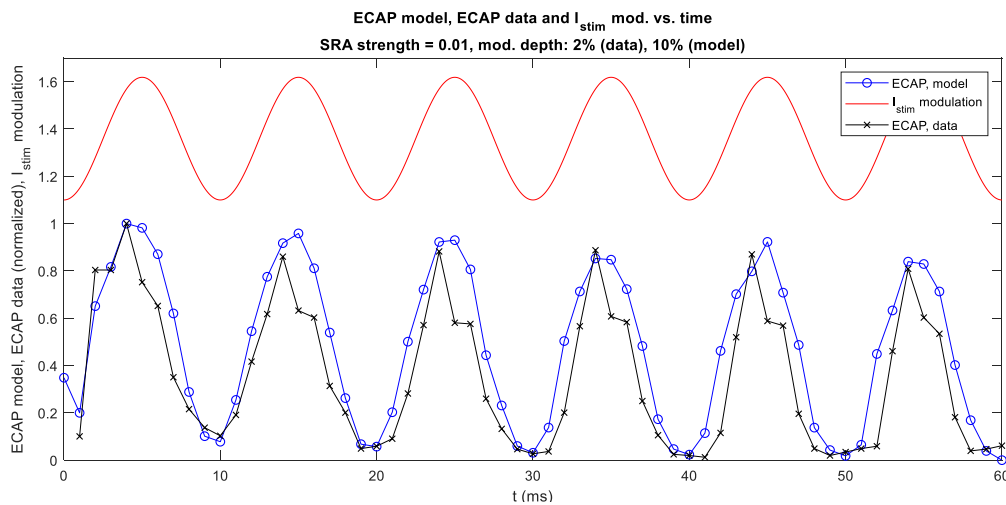


Figure 4.8. Predicted and observed normalized ECAP response for sinusoidally modulated pulse train, with modulation envelope shown above with arbitrary offset. Same parameters as in Fig. 4.7. except that the stimulus modulation depth has been increased to 10%.

To test our predictions from above, SRA strength was increased from 0.01 to 0.03, which produced the response as shown in Fig. 4.9. below, where the reduction in response width with can be noticed as well as a slight phase shift. As the response modulation depth was still somewhat low compared to the data (especially at stimulus maxima in later pulses), the stimulus modulation depth was further increased to 8%; the resulting response can be seen in Fig. 4.10 below, which now matches the data ECAP response more closely. The tentative reason for increasing the modulation depth as compared to the square modulated pulse trains is that the stimulus level of sinusoidally modulated pulse trains is higher, and since the slope of AGF is expected to become somewhat higher for higher stimulus levels, this would produce a higher response modulation depth.

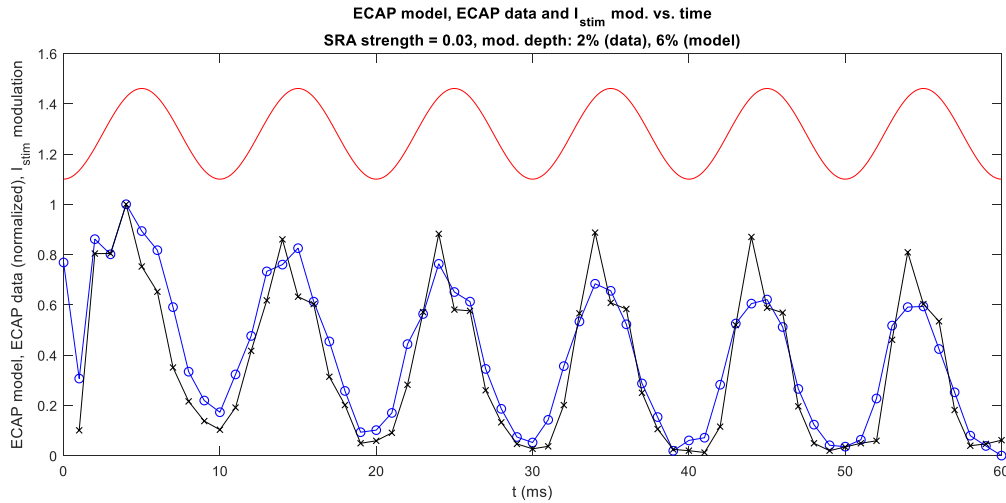


Figure 4.9. Predicted and observed normalized ECAP response for sinusoidally modulated pulse train, with modulation envelope shown above with arbitrary offset. Same parameters as in Fig. 4.7. except that the spike rate adaptation strength has been increased from 0.01 to 0.03.

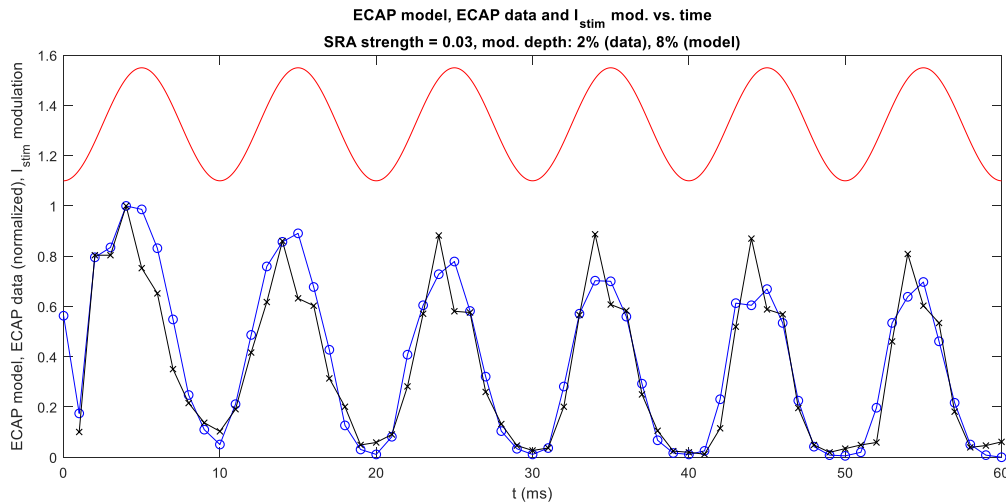


Figure 4.10. Predicted and observed normalized ECAP response for sinusoidally modulated pulse train, with modulation envelope shown above with arbitrary offset. Same parameters as in Fig. 4.9. except that the modulation depth has been increased from 6% to 8%.

4.5. Modulated pulse trains: saw modulation

Finally, we shall consider the responses to saw modulated pulse trains; although they are a somewhat unrealistic modulation type to be “observed” in a cochlear implant, a previous study by [Hess 2017] has identified an interesting phenomenon observed in ECAP response to saw modulated pulse trains which we shall attempt to explain.

As can be seen in Fig. 4.11. below, the observed ECAP response shows an ever later onset and an increasing slope with each stimulus period. Such behavior can be again correlated with an ongoing reduction in neural excitability, which effectively increases the threshold for each stimulus period, thus delaying the onset of ECAP response. Without the effect of spike rate adaptation being active, as seen in modelled ECAP response in Fig. 4.11., the predicted ECAP response exhibits a constant slope during the stimulus and thus does not capture the observed response dynamics.

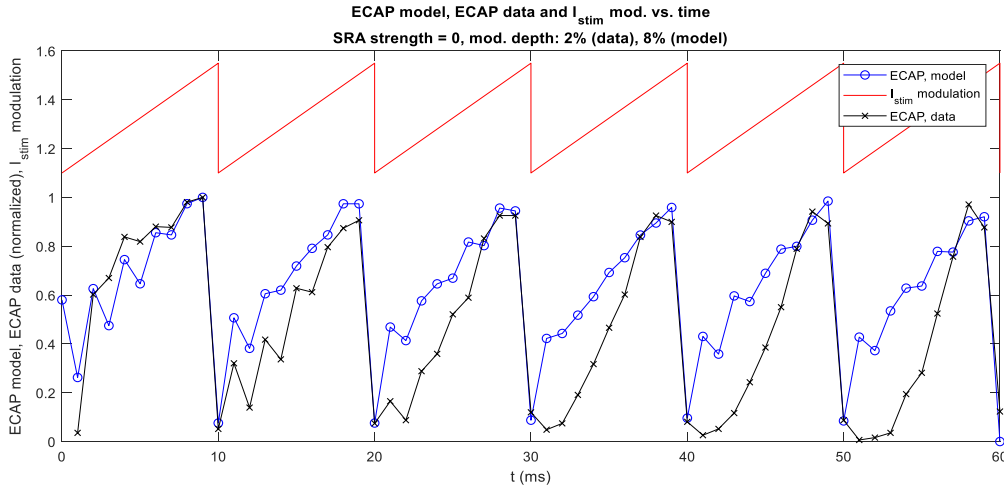


Figure 4.11. Predicted and observed normalized ECAP response for saw modulated pulse train, with modulation envelope shown above with arbitrary offset. Same parameters as in Fig. 4.10. except that SRA has been deactivated (modulation depth is 8% and maximum stimulus level is 225 CL).

However, by increasing the SRA strength sufficiently (Fig. 4.12. and Fig. 4.13.), this general behavior can be captured by our model, though yet again, the maximum amplitudes reached in later stimulus periods are systematically lower (as observed also in other modulation types) and could not be completely matched by lowering the accommodation strength. Since this implies that the effect of SRA is too strong, one should reduce the SRA strength, but this will lead to reduced accuracy in fitting the late onset and increased slope observed in saw modulated pulse trains. On the other hand, increased modulation depth would help produce the late onset effect, but as previously, the appropriate amount of modulation depth depends on resolving the discrepancy in modelled and observed AGF.

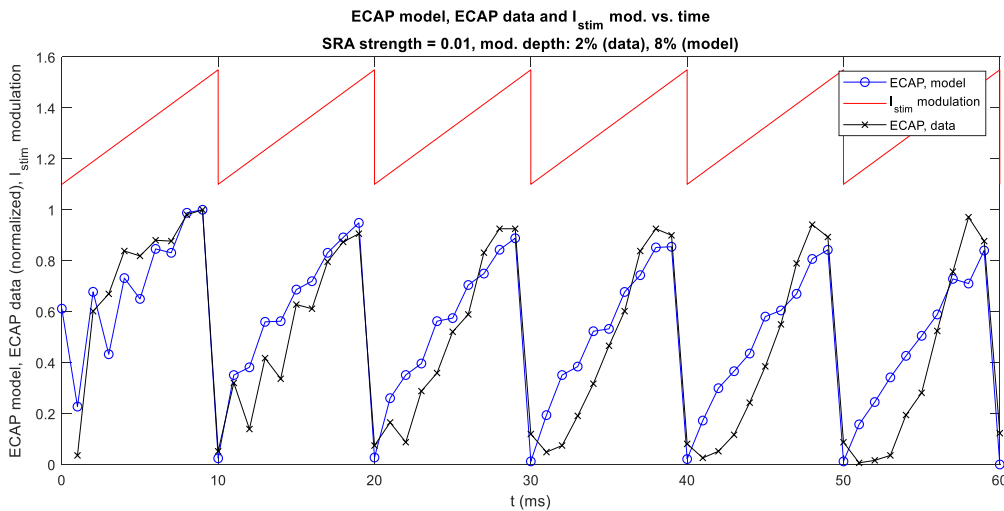


Figure 4.12. Predicted and observed normalized ECAP response for saw modulated pulse train, with modulation envelope shown above with arbitrary offset. Same parameters as in Fig. 4.11. except that SRA strength has been increased to 0.01.

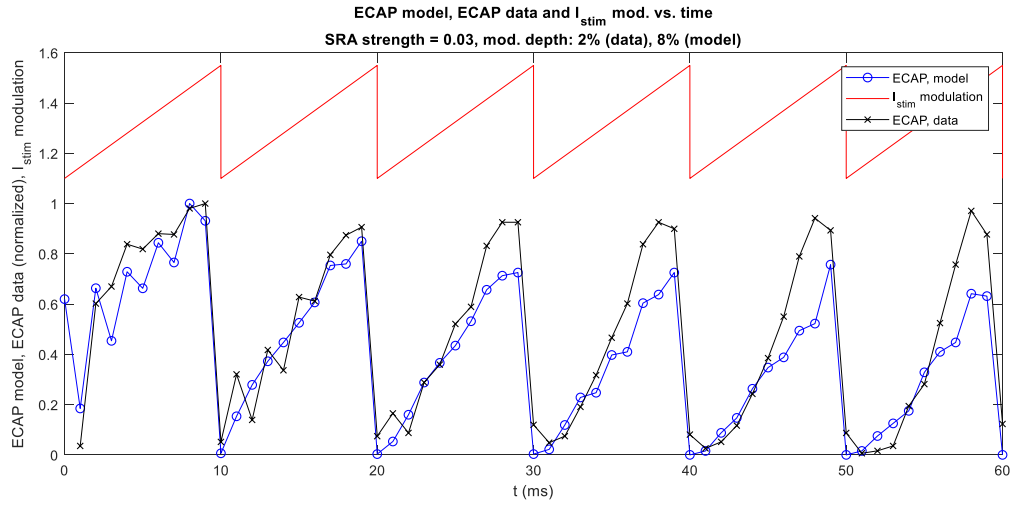


Figure 4.13. Predicted and observed normalized ECAP response for saw modulated pulse train, with modulation envelope shown above with arbitrary offset. Same parameters as in Fig. 4.11. except that SRA strength has been increased to 0.03.

Though not presented here, the previous study by [Hess 2017] has also considered responses to reverse saw modulated pulse trains: these show an inverse effect (relative to saw modulation) of increase in ECAP response slope and an ever sooner decrease of ECAP response to noise level, which can be attributed again to spike rate adaptation effects.

5 Conclusion

In this study, a single neuron phenomenological neural model by [Boulet et al 2018] has been extended to a population of neurons, and in addition to the already included effects of noisiness, refractoriness, spike rate adaptation, facilitation and accommodation, the effects of spread of excitation, variability in refractory time scales (ARP/RRP durations) have been introduced based on [Hamacher 2004].

Moreover, to enable easier testing of model predictions for potential users, a graphical user interface has been developed which allows arbitrary combinations of the above mentioned neural phenomena, modifications of all relevant neural and stimulus phenomena, and saving of any data or figures calculated in this model. To enable comparison of model predictions to NRT data an ECAP model was implemented and used to calculate the predicted ECAP responses using the calculated spike histogram, as well as to calculate “synthetic” amplitude growth function, recovery function and facilitation function.

From the computational side, implementation of this code shows a reduced computational load (shorter execution time) compared to a similar model by [Tabibi et al 2018], and further modifications which can lead to further acceleration of code runtime have been identified, such as using the midpoint method for solving differential equations in the model, and the possibility of performing massively parallelized computations on GPU, given that the single remaining memory bottleneck is resolved (tentatively in the manner proposed in this study).

Advantages of this model are the ability to match the observed behaviors ECAP responses to various modulated pulse trains, including: oscillations for square modulation, phase shift and reduction in width for sinusoidal modulation, and the increase/decrease in slope and onset of response for (reverse) saw modulated pulse trains. In the majority of these cases, the dominant effect responsible for the observed behavior was the spike rate adaptation which causes an ongoing decrease in excitability with sustained spiking. However, the value of SRA strength and (model) modulation depth had to be adjusted between various cases in order to fit the data, thus reducing the predictive power of our model, and it is expected that resolving the discrepancy in modelled versus data modulation depth would resolve this issue.

This key deficiency of our model lies in the simplistic approach used to calculate the ECAP response which, contrary to results observed with CI patients, predicts the objective (NRT) threshold to be equal to the subjective threshold of a patient. This is caused by the assumption that practically no latency is present in individual action potentials constituting the ECAP response, and the practical implications of this are that the slope of AGF is lowered, which produces a lowered modulation depth in the predicted ECAP response. Alleviating this shortcoming has been performed in an „ad hoc“ manner, where stimulus modulation depth in the model was increased e.g. from 2% in the data to 6% in the model, which has produced satisfactory results.

In future work based on this study, attention should be directed in improving the ECAP model used, the „weakest link“ in our compound model, since the current model prevents proper comparison of predicted and measured ECAP responses and thus proper validation and usage of our model. Furthermore, since the exact shape and (extrapolated) threshold of amplitude growth function depend on the neural survival and membrane threshold, which have variable values across different neural fibers, it is advised to include these effects in future versions of this model. Tentative methods to obtain these parameters from ECAP data have been proposed in this study, and it is expected that these parameters will provide useful information such as the presence of regions with low neural survival which could then be avoided during stimulus generation for a cochlear implant.

Likewise, measurements needed to extract other parameters for our model have been suggested, and due to a limited set of data used in this study, it is expected that such measurements will provide a more suitable test of our model. This will hopefully help reach a verdict on the usability of our model for predicting responses of neural populations to stimuli generated by novel coding strategies aimed at improving hearing for cochlear implant patients, which was the main goal of this study.

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